Deepfake Image Detection Using Self-Supervised Contrastive Learning and Slot Attention

Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in Computer Science and Engineering of the University of Asia Pacific

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Declaration

We, hereby, declare that the work presented in this Thesis is the outcome of the investigation performed by us under the supervision of Nakiba Nuren Rahman, Lecturer, Department of Computer Science and Engineering, University of Asia Pacific. We also declare that no part of this Thesis and thereof has been or is being submitted elsewhere for the award of any degree or Diploma.

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We hereby recommend that the thesis prepared by Mahia Akter Momo, Sumaya Jahan Othye, and Nafisa Ali entitled "Deepfake Image Detection Using Self-Supervised Contrastive Learning and Slot Attention" is accepted as fulfilling the part of the requirements for degree of Bachelor of Science in Computer Science and Engineering.

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Abstract

The fast development and proliferation of deepfake technology presents grave challenges to data integrity, privacy and digital security. While traditional supervised deep learning models have demonstrated some effectiveness in deepfake identification, they may be prone to overfitting on background textures and not perform well on new manipulation techniques. This thesis aims to solve these problems by proposing a comprehensive deep learning model for deepfake image detection combining a Slot Attention mechanism with self-supervised contrastive learning. The proposed method consists of two main steps: In the first step, a ResNet inspired convolutional encoder is pre-trained along with an 8 slot Slot Attention layer using self-supervised contrastive learning and Normalised Temperature scaled Cross Entropy (NT-Xent) loss. This architecture allows the model to automatically break down the features into object centric representations, which makes it possible to detect errors that are introduced due to the changes in the local features and facial inconsistencies without any manual annotation. Second, a supervised classification head is attached to the trained encoder and fine tuned to perform a binary classification task for real or fake. The proposed framework is evaluated on the OpenDeepfake Preview dataset, which consists of 19,999 images, and the test accuracy of the proposed framework is 90.73%, which is far superior to the test accuracy of 88.0% of the traditional supervised CNN baseline. Experimental results prove the benefits of the combination of self-supervised pre-training and slot based localized attention for better detecting signs of tampering and enable a highly robust and generalizable approach for verifying the authenticity of digital media.

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Chapter 1

Introduction

The research context, motivation and tentative scope of this thesis project is discussed in the introduction of this chapter. It begins by laying out the status of deepfake generation technology and all the constraints that have existed on existing detection approaches and highlights the need for self-supervised contrastive learning and object centric attention. The exact problem is then conceptualized along with the practical goals, research questions and hypotheses of the project. Finally, the outcomes of the expected, the broader social, legal and environmental impacts and the structure of the report are summarised.

1.1 Background and Motivation

In recent years, the ability of Artificial Intelligence (AI) and deep generative models has revolutionised the synthesis of media. In the case of these technologies (such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs) and Diffusion Models (DMs)), a highly realistic media of synthetic images can be obtained, also known as deepfakes. Digital security, privacy and trust in the public is in danger by innovative technologies for entertainment, education and digital art. The deepfakes can be employed to generate pornography without permission, to spread political rumors, to steal identities and to conduct monetary cons. Thus, the creation of strong, automated and transferable deepfake detection methods is a pressing concern to ensure the integrity of the digital communication system.

The majority of traditional deepfake detection methods have been based on fully supervised learning methods that are trained with a set of real and fake images that have been labeled. These models are good at performing on the data they are trained upon, but they have numerous limitations in terms of generalisations. In particular, supervised models tend to overfit on artifacts, background textures, compression rates or post processing footprints in the given set of data. Their performance decreases significantly when they come across unseen deepfake generation approaches or out of distribution samples. The restriction is that supervised training is done on the shortcut features (high frequency noise patterns, background characteristics etc.) and not on generic facial semantics.

As an alternative, self-supervised representation learning has become a promising approach that gets rid of such drawbacks. The model is pretrained with unlabeled data by contrastive learning to learn general representations of facial shape, structure and boundary information of texture. Moreover, an object centric attention mechanism, such as Slot Attention, can be introduced to divide feature maps into localized and interpretable regions (slots) within the map. This way, the detector can focus on the inconsistent areas, or the manipulation regions in the local area, which are around the eyes, nose and mouth, rather than on irrelevant background textures. Based on the possibility of self-supervised pretraining and slot based localized attention, this research aims to develop a powerful and general deepfake detection system.

1.2 Problem Statement

Despite growing numbers of deepfake detection algorithms, existing models still face various limitations. Supervised detection models are highly domain-shift sensitive and overfit to high frequency noise and background textures in their training dataset. Thus, the performance of these models drops when testing on deepfakes created using newer or unknown architecture. Moreover, convolutional neural networks are prone to generate global features that ignore forgery localization cues. Considering that manipulations in deepfakes occur within certain parts of faces like the eyes, lips, or blending area, such models may be fooled by the untouched background. Finally, supervised detection methods depend greatly on accurately annotated examples of both real and fake faces, making them highly dependent

on labeling process and inflexible to novel manipulations.

Thus, there arises a need for developing a deepfake detection system capable of learning feature representations of faces without extensive annotations and able to separate the areas of manipulations from the background.

1.3 Objectives of the Project

This thesis proposes the following primary goal to be achieved through designing and implementation of a custom framework for robust deepfake detection, namely, a combination of a self-supervised contrastive pretraining method with the use of Slot Attention mechanism. First, the ResNet architecture based convolutional encoder is trained with a self-supervised contrastive learning algorithm using the NT-Xent loss function in order to extract generalized representations of facial structures, shapes, and contours using unsupervised approach. Then, the use of an 8-slot Slot Attention allows partitioning feature maps into the object centric representation in order to separate facial regions and extract local forgery artifacts. After that, the binary classification task between the real and fake images is accomplished by training a classification head on the pretrained slot based representation. Finally, the efficacy of the proposed framework is tested on the OpenDeepfake Preview dataset containing 19,999 images and its results are compared with those of a supervised CNN classifier on the basis of the evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

1.4 Research Questions / Hypotheses

This study aims to be able to answer the following research questions:

- **RQ1:** How well can self-supervised contrastive pretraining enhance the feature representation of facial textures compared with supervised learning directly?
- **RQ2:** How well can a Slot Attention module localize and isolate manipulated facial parts in a background?
- **RQ3:** Is it possible for the proposed self-supervised Slot Attention mechanism to

outperform a traditional supervised CNN baseline model in terms of generalization and accuracy?

Based on these questions, the following hypotheses are defined:

- **Alternative Hypothesis (H1):** The proposed self-supervised Slot Attention framework is found to be statistically significantly more accurate and robust in classification when compared to the traditional supervised CNN baseline.
- **Alternative Hypothesis (H1):** The proposed self-supervised Slot Attention framework is significant to improve classification accuracy and robustness due to the separation of localized manipulation artifacts by the self-supervised pretraining and attention mechanism.

1.5 Scope of the Work

There are various limitations to this project. The temporal analysis of deepfake videos cannot be conducted through this research; the research will only cover the detection of deepfakes in images. The model used in this project is based on the OpenDeepfake Preview dataset, which consists of 19,999 images, which have been evenly divided into two classes, namely real and fake images. The encoder used in this project consists of a custom built Convolutional Neural Network ResNet having a structure of five layers of residual convolutions. The Slot Attention layer has been used in this project, which has 8 slots and features dimension of 512. Besides, this project is limited by the fact that it involves training and testing the model on the local GPU (NVIDIA MX350) and cloud based GPUs (Kaggle & Google Colab:NVIDIA T4 GPU).

1.6 Expected Outcomes

At the end of this project the following outputs and outcomes are expected:

- End to end deep learning solution to classify images as deepfake.

- Highly robust, generalizable facial features pre trained encoder weights.
- An interpretable Slot Attention layer for generating spatial attention maps, which are used to determine which facial regions the model detected as manipulated.
- A thorough evaluation report shows that the proposed model has achieved around 90.73% accuracy in the test phase, which is quite higher as compared to 88.00% accuracy of the supervised baseline.

1.7 Impacts of the Project

The value of the development and implementation of this deepfake detection framework lie in the technical and social value. The project will help to overcome major shortcomings in the modern information network, such as the authenticity of digital media. It serves as a barrier to the malicious use of generative AI, helping to keep the integrity of public communication channels and maintaining digital trust.

1.7.1 Impact on Societal and Cultural Issues

Deepfakes have a devastating impact on society because they deceive viewers of images. They can be used to influence democratic elections, for the dissemination of fake news and the defamation of a person who is disproportionately targeted by the marginalized groups and by non consensual synthetic media. This study provides a more generalizable detection framework to help combat digital deception. It helps rebuild a sense of trust in the media institutions and it also offers cultural protection by providing a means to verify authenticity of sensitive historical, political and cultural records.

1.7.2 Impact on Health, Safety, and Legal Issues

Deepfakes can also be used for identity theft and financial fraud, posing a significant security threat from a safety and legal standpoint. Our model is important in legal proceedings in determining the admissibility of digital evidence and is also used as a forensic tool to test the

authenticity of photographic evidence. Deep generative models could be applied to create new medical images (such as adding or removing lesions in computed tomography [CT] scans) in healthcare. The proposed system can be adapted to authenticate biomedical data, ensuring patient safety. Moreover, our framework follows rigorous data privacy regulations to avoid legal and security concerns while deployment, making sure that images utilized by the user throughout the inference are not saved or abused.

1.7.3 Impact on Environment and Sustainability Issues

Exercising these gigantic and potent deep learning models demands a significant number of compute cycles, which consequently causes carbon emissions to be released. The project will focus on the pipeline to load image as required making it less complex in terms of training and taking much lesser RAM thus catering to Environmental sustainability. Finally a light weight, little space used Slot Attention layer on top of a ResNet style backbone: squeezing the params to as low as $\approx$ 2.26 million so we won't need to waste thousands of hours of GPU compute training huge Vision Transformers. As AI is the next big technology trend out there, because training a model is expensive and this design can reduce carbon footprint (both in training and inference) which helps to sustainable development of Artificial intelligence field.

1.8 Report Organization

The thesis is comprised of eight chapters. The introduction to the research topic is given in chapter 1, including the background of the study, problem statement, research objectives, scope, limitations, and motivation of the study. Background and literature review is given in chapter 2, wherein the discussion on deep fake generation and detection, approaches, developments, and literature review have been conducted in order to identify a research gap. System analysis and design is provided in chapter 3, wherein the discussion on the requirements, architecture, workflow, and other design aspects of the proposed framework is conducted. Methodology and implementation of the proposed research is introduced in chapter 4, wherein the discussion on the dataset, pre-processing method, self-supervised

contrastive learning approach, slot attention approach, architecture, training, and implementation of the proposed framework is provided. Results of the experiments performed and performance of the proposed framework with the help of standard metrics are provided in chapter 5. Comparative analysis of the proposed framework is conducted using the baseline CNN model. The plan of action, schedule, and budget required for the research project are discussed in chapter 6. Engineering standard and design constraint is discussed in chapter 7.Conclusions of the thesis are presented in Chapter 8.

Chapter 2

Background and Literature Review

In this chapter, the theory of deepfakes and the latest research related to the detection of deepfakes in images will be summarized. It begins by laying down the mathematical basics of Self-Supervised Contrastive Learning and Slot Attention. It then carries out a detailed literature review, which is divided into two areas: contrastive based face forgery detection and object centric visual representation learning. At the end, a comparative study is conducted to specify the fundamental research gaps and limitations of the current methods.

2.1 Preliminaries

Image forensics based on deepfakes entails determining if a facial image is manipulated through face swapping, reenactment, editing, and generating synthetic faces [1]. This task poses a lot of difficulty owing to the current sophisticated generative techniques—such as Generative Adversarial Networks (GANs) [2], StyleGAN architectures [3], Variational Autoencoders (VAEs) [4], Denoising Diffusion Probabilistic Models (DDPMs) [5], and Latent Diffusion Models [6]—that make deepfake images visually realistic; and also the manipulations might only affect a localized region on the face. Therefore, the task requires a forensics model that learns visual representations that detect the artifacts of manipulation without getting affected by irrelevant changes like the pose, illumination, compression, and background noise. Traditional supervised models often leverage deep pre-trained backbones like Xception [7] or MesoNet [8] for spatial anomaly detection [9,10], but their generalization remains limited. Representation learning models have therefore been proposed in recent

research.

2.1.1 Self-Supervised Contrastive Learning

Contrastive self-supervised learning is another approach to representation learning which does not require any labeling for training at all. Here, the positive pairs consist of different augmentations of the same picture while the negative pairs consist of different pictures being augmented. Training is performed so that the embeddings of the positive pairs get closer to each other and negative pairs get farther away from each other. In the case of the SimCLR framework [11], two augmentations are performed on the input picture after which the outputs pass through the encoder and the projection head, and are then optimized via NT-Xent loss. Other self-supervised contrastive learning paradigms, such as Momentum Contrast (MoCo) [12–14], bootstrap-based representation without negative pairs (BYOL) [15], clustering-based representation (SwAV) [16], and redundancy reduction models like Barlow Twins [17], have also demonstrated robust feature learning without human annotations. Contrastive self-supervised learning is very important since it is helpful in getting the structure and meaning of the images instead of remembering labels. More specifically, x_i and x_j are considered to be two augmentations of the same picture. The features are extracted using the encoder function $f(\cdot)$ while projections are made using the projection head function $g(\cdot)$. NT-Xent loss is used to calculate the distance between the positive pairs and other pairs. Scaling of temperature affects the model's ability to learn hard negative samples, whereas cosine similarity is used as a measure of distance between two vectors in the embedding space. From the standpoint of solving the deepfake recognition task, the relevance of the proposed architecture is related to the fact that the model is able to extract inhomogeneous texture, borders, and high frequency features in an unsupervised manner. The advantages of contrastive self-supervised learning include the possibility of using unlabeled data for training in case of a lack of labeled deepfakes or new deepfake generation techniques. DeepfakeUCL [18] demonstrates the effectiveness of contrastive learning in deepfake recognition without supervision on the dataset as well as cross dataset. Contrastive learning was further improved with the addition of the supervised part responsible for separation of classes and generalization in case of domain shift [19–21]. With respect to my research, the application of self-supervised contrastive learning can be

a good approach for the extraction of deepfake features.

2.1.2 Slot Attention Mechanism

Slot Attention [22] is an object oriented representation technique where the input is divided into several latent slots which are meant to represent separate elements or regions of interest. In contrast to global pooling, Slot Attention takes advantage of competition among the attention heads to assign input features into a predetermined number of slots allowing the network to represent disjointed parts of the scene. Aimed at providing a capability for compositional reasoning and generalization to new object configurations, Slot Attention was shown to be capable of capturing object-like representations from the visual input [22]. This object-centric perspective has since been extended to various domains, including scene parsing [23], video object segmentation [24], and spatial priors [25]. In the framework of Slot Attention [22], the feature map is projected into keys and values while the slots' vectors act as queries. Attention scores are calculated between slots and input tokens and the slot updates are achieved through iterative computation of the weighted sum of the input with subsequent refinement of the results. Due to competitive normalization in slots, a large amount of input features is assigned to a particular slot helping to split the input regions which might be overlapping or entangled. This characteristic is especially relevant to the problem of deepfake detection since the areas which have been altered in manipulations are typically located close to the eyes, mouth, cheeks or borders, while other parts of the image have not undergone any alterations. Taking into account the aforementioned characteristics, Slot Attention can be applied in the context of the proposed thesis framework to segment facial subregions after the process of feature extraction and reduce dependency on the background context. It is a very important aspect, as many detectors using global embeddings tend to overfit to the specific features of the dataset instead of forgery related features.

2.2 Related Works / Existing Systems

The literature relating to this research problem topic, as reviewed in general surveys [1, 26], may be grouped into two main streams: deepfake detection based on contrastive learning

[18–21, 27–32] and object focused visual representation learning [23–25, 33–37]. Both streams complement each other; the former increases the quality of the features' learning, whereas the latter fosters their spatial decomposition.

2.2.1 Contrastive Learning in Deepfake Detection

One of the earlier works in which contrastive learning was applied for the task of deepfake detection was DeepfakeUCL [18]. The idea of using contrastive learning in an unsupervised way for face forgery detection was realized. The approach involved creation of multiple views of the same image and learning of features vectors which did not depend on the applied augmentation. The key contribution of this paper was in the possibility to learn the informative forgery specific features without using class labels assigned by humans. Subsequent studies made use of contrastive framework to improve generalization capabilities for the practical scenario, testing on benchmark datasets such as FaceForensics++ [38] and other forgery sets. The use of supervised or hybrid contrastive framework was made in order to provide better distinction between real and fake classes while preserving the robustness across multiple datasets [19–21, 27, 29–31]. Contrastive learning frameworks making use of multi scale RGB frequency features [32] make use of the spatial and frequency domain features to expose the manipulation artifacts that would go unnoticed in the RGB space. These techniques are inspired by the fact that deepfakes often involve small discrepancies in terms of texture, blending, and frequency distribution [39]. On the other hand, there are several approaches that deal with the difficult cases including video forgery detection and completely unsupervised detection. The models that use hierarchical contrastive inconsistency learning [28] take advantage of the spatial temporal inconsistencies in videos, while fully unsupervised detectors of deepfakes [31] employ pseudo labeling and improved contrastive learners. While all these models demonstrate good results within the scope of their applicability, most of them rely on the global embedding and hence suffer from the same limitations.

2.2.2 Object-Centric Visual Representation Learning

The construction of an object oriented representation includes the decomposition of the image into several structured latent elements as opposed to the encoding of the entire image into a single vector. One of the most famous architectures of such networks is the Slot Attention network [22] due to the ability of the architecture to learn several latent slots which correspond to the related objects or components of the scene. This property is necessary both for the purpose of interpretability and localization, because each latent slot corresponds to a particular part of the visual scene. According to the original paper about Slot Attention [22], the property of the proposed architecture to learn object centered representations and extrapolate new combinations is quite important for the purposes of the current research, because the deepfakes are typically compositional and localized. In other words, different components of faces, such as eyes, mouth contours, or any other components of the face, are able to appear together with untouched background or skin. This way, the detection of manipulated areas becomes simpler than in the case of global representation learning. Still, there are several works that have shed light on the advantages of object representation in visual reasoning. Although these papers do not refer to any deepfakes at all, they prove the efficiency of the slot based methods in constructing the scene graph, robot memory, and visual perception in terms of actions [23, 24, 33–37]. What makes these studies relevant for the present study is the following: the slot based methods can help to identify the crucial objects among the irrelevant backgrounds. Slot Attention looks very relevant for the forensic image analysis, even though it is seldom used for the deepfake images.

Table 2.1: Comparative Analysis of Related Works

Research Year	**Research Focus**	**Dataset**	**Modeling Techniques**	**Performance**	**Limitation**
Larue et al., 2023 [27] *"Seeable: Soft discrepancies and bounded contrastive learning for exposing deepfakes" (ICCV 2023)*	Unsupervised anomaly-based deepfake detection	FaceForensics++ (FF++), Celeb-DF (V2)	Anomaly detection framework utilizing synthesized "soft discrepancies" on real faces; bounded contrastive loss under facial geometric prior constraints.	Strong generalization on unseen manipulation techniques without training on fake generators.	High false-positive rate on real images containing compression noise or natural skin blemishes; depends on rigid facial geometric templates.
Sun et al., 2023 [31] *"Contrastive pseudo learning for open world deepfake attribution" (ICCV 2023)*	Open-world deepfake source attribution	FaceForensics++ (FF++), Celeb-DF, and model-specific subsets	Contrastive Pseudo Learning (CPL) for semi-supervised clustering to distinguish known and unknown generator fingerprints.	High classification precision in identifying both known and novel generator architectures.	Overconfidence bias in pseudo-labeling known classes degrades attribution accuracy on novel attacks; assumes number of unknown classes is known a priori.

Research Year	Research Focus	Dataset	Modeling Techniques	Performance	Limitation
Dong et al., 2023 [32] *"Contrastive learning-based general deepfake detection with multi-scale rgb frequency clues" (JVCIR 2023)*	General deepfake detection	Face-Forensics++ (FF++), Celeb-DF	Dual-modality contrastive learning combining spatial RGB representations and multi-scale frequency clues (DCT).	Highly accurate on high-frequency forgery artifacts under intra-dataset evaluations.	Performance drops significantly under standard image degradations (e.g., JPEG compression or downsampling) which erase frequency clues.
Wu et al., 2023 [25] *"Slot attention with spatial prior for object-centric representation" (CVIU 2023)*	Unsupervised object-centric scene segmentation	CLEVR-based multi-object sets, synthetic scenes	Slot Attention modified with coordinate-based spatial priors for anchoring query/slot initialization.	Outperforms standard Slot Attention in unsupervised segmentation on cluttered backgrounds.	Rigid spatial priors can fail to capture objects that deviate from typical spatial distributions; not tested or adapted for facial geometries.

Research Year	Research Focus	Dataset	Modeling Techniques	Performance	Limitation
Zheng et al., 2024 [19] *"Deepfake detection with combined unsupervised-supervised contrastive learning" (ICME 2024)*	Generalizable deepfake detection	FaceForensics++ (FF++), Celeb-DF (V2)	Hybrid framework combining unsupervised contrastive pre-training with supervised classification.	Improved cross-dataset generalization compared to purely supervised classifiers.	Overfits to static, frame-level noise patterns and local textures; vulnerable to domain shift from diffusion models.
Hong et al., 2024 [30] *"Contrastive learning for deepfake classification and localization via multi-label ranking" (CVPR 2024)*	Localization-aware deepfake classification	FaceForensics++ (FF++), Celeb-DF	Contrastive learning combined with multi-label ranking loss to map and localize manipulated segments.	Successfully localizes manipulated regions (eyes, mouth) while performing image classification.	High computational complexity; modeling the sequence of stacked manipulations is complex and error-prone.
Didolkar et al., 2024 [36] *"Object-centric representation learning for complex real-world scenes" (Neural Networks 2024)*	Object-centric representation in real-world scenes	COCO, PASCAL VOC	Slot Attention combined with self-supervised foundation features (DINO) for natural scenes.	Successfully scales unsupervised object discovery to complex real-world datasets.	Requires a fixed, pre-defined number of slots (K); foreground details sometimes bleed into background slots.

Research Year	Research Focus	Dataset	Modeling Techniques	Performance	Limitation
Son & Kim, 2025 [21] *"Advancing generalization in deepfake detection: Supervised contrastive representation learning with dual stream spatio-temporal features" (IEEE Access 2025)*	Generalization in deepfake video detection	Face-Forensics++ (FF++), Celeb-DF	Supervised contrastive representation learning with a dual-stream (spatio-temporal) framework.	High robustness and generalization in video cross-dataset evaluations.	High computational latency of dual-stream networks; cannot be applied to single compressed static images.
Jiang et al., 2025 [37] *"Slot attention networks for medical image segmentation" (MedIA 2025)*	Interpretability in medical image segmentation	Computed Tomography (CT), MRI datasets	Multiscale Slot Attention networks combined with convolutional encoders.	Competitive segmentation of organs/lesions with interpretable slot attention regions.	Struggles to resolve vaguely defined pathological boundaries and extreme scale variations.

Research Year	Research Focus	Dataset	Modeling Techniques	Performance	Limitation
Chung et al., 2026 [33] *"Rethinking progression of memory state in robotic manipulation: An object-centric perspective" (ICRA/AAAI 2026)*	Memory-state modeling in robotic manipulation	LIBERO-Mem manipulation benchmark	Object-centric State Space Model (SSM) with Gated Recurrent Unit slot updates.	Successfully tracks long-term object state dependencies in robotic manipulation.	Naive implementation depends on oracle subgoal representations; validation is confined to simulation.
Le et al., 2026 [35] *"Uno: Unifying one-stage video scene graph generation via object-centric visual representation learning" (CVPR 2026)*	Video scene graph generation (VidSGG)	Action Genome, Volumetric Video datasets	UNO: Unified One-stage scene graph generation via object-centric slots.	High tracking and relationship-mapping accuracy across dynamic scenes.	Fixed slot capacity limits performance in highly crowded or fast-motion environments.

The comparative analysis indicates that recent deepfake detection research has progressively shifted from conventional supervised learning toward contrastive and self-supervised learning to improve robustness and cross-dataset generalization. Studies by Larue et al. [27], Dong et al. [32], Zheng et al. [19], Hong et al. [30], and Son and Kim [21] reported strong detection performance by learning discriminative feature representations through

contrastive objectives. However, many existing methods rely on manipulation of specific artifacts, frequency domain features, temporal consistency, or large amounts of labeled data, limiting their effectiveness under image compression, domain shifts, and unseen deepfake generation techniques. Meanwhile, Slot Attention [22] has shown remarkable success in object-centric representation learning for scene understanding, medical image segmentation, and robotic perception by capturing meaningful object-level features. Nevertheless, these approaches have not been specifically designed or evaluated for facial deepfake detection and often require predefined slot configurations or struggle to adapt to subtle facial manipulations. Overall, the reviewed studies reveal a clear research gap in integrating self-supervised contrastive representation learning with object-centric feature extraction for generalized deepfake image detection. To address this limitation, the proposed framework combines contrastive self-supervised pretraining, Slot Attention, and supervised fine-tuning to learn robust and semantically meaningful facial representations. This integrated approach is expected to improve robustness, interpretability, and generalization across diverse datasets and previously unseen deepfake manipulation techniques.

2.3 Research Gap / Limitations of Existing Methods

From the reviewed literature, two major gaps have been found that have served as motivation for this thesis work. The first one is the usage of global embeddings by the vast majority of contrastive learning based deepfakes detection algorithms [18, 19, 32], which leads to a risk of the model using dataset specific background information or compression/camera noise rather than forgery specific facial features. Consequently, the quality of the algorithm may deteriorate when used on new datasets or on images acquired under other circumstances.

The second gap is that localization is a passive process in contrastive learning based models, as some papers use techniques like attention visualization, ranking loss [30], or post hoc explainability methods [20]. However, in contrast to the aforementioned techniques, Slot Attention [22] forces the encoder to split the image into several slots, thus making it actively separate different facial parts. This is crucial since the forgery is usually localized, while the background is intact.

An additional drawback within the existing literature is the lack of any object oriented model for deepfakes. Although Slot Attention [22] has been shown to be highly efficient for generic object detection and scene decomposition tasks [22, 36], its use for detecting forged images is yet to become common practice. This can easily become one of the areas of research, in which case a combination of self-supervised contrastive pre-training with the use of Slot Attention may prove helpful in generalization and localization tasks. In the context of the proposed thesis, this will be achieved through feature learning and then slotting.

2.4 Summary

In this chapter, the theoretical background and the review of related literature have been discussed for the deepfake image detection framework proposed in this paper. Self-supervised contrastive learning can be considered as an efficient technique for acquiring robust features without any supervision, while Slot Attention can be utilized as an object centric technique.As a result of the review of existing works, it is clear that although contrastive techniques are effective,while object centric techniques are structured and not much utilized in deepfake detection.

Chapter 3

System Analysis and Design

The requirement analysis and the architectural design of the suggested deepfake image detection framework are introduced in this chapter. It explains functional and non-functional requirements, explains the high level and detailed architectural configurations, explains the design rationales for other configurations, and gives the full mathematical formulations and pseudocode for the training phases of the model.

3.1 Requirements Analysis

The needs and requirements of building a strong and generalizable deepfake detection system are grouped into functional requirements (what the system should be able to do) and non-functional requirements (the boundaries and performance expectations of the system).

3.1.1 Functional Requirements(FR)

- **FR1: Image Pre-processing and Pipeline loading**

  The system needs to preprocess the images from the balanced OpenDeepfake Preview dataset, scaling the images to a size that is dynamic depending on the MX350 environment (128×128 for local MX350 and 224×224 for cloud based T4 GPU environments), and rescaling the pixel intensities back into the $[0, 1]$ range.

- **FR2: Contrastive Learning with Data Augmentation**

  To form positive pairs to be used in self supervised learning, a pre-processing pipeline

should produce two different stochastic and semantically preserved views of each input face image by applying random flipping, random rotations, random brightness and contrast adjustments.

- **FR3: Feature Extraction**

  The system should be based on a convolutional encoder to obtain high dimensional spatial feature representations of the input images provides a feature partitioning approach based on objects. The architecture should include a Slot Attention layer with 8 slots which is capable of dynamically grouping the spatial feature maps into discrete, localized, object centric representations.

- **FR5: Binary Classification**

  The framework should be trained to map the slot representations to a dense classification head, which would then produce a probability score, between 1 and 0, whether the input face is real or fake.

- **FR6: Forensic Localization Outputs**

  The system should be able to create spatial attention maps from Slot Attention layers to indicate the region of face segments that resulted in forgery detection.

3.1.2 Non-Functional Requirements

- **NFR: High Generalization and Accuracy**

  The system should reach at least 90% of accuracy on a classification test, with better performance than the traditional fully supervised baselines (88.00%) and robustness against the texture overfitting problem.

- **NFR2: Memory and Resource Constraints**

  The data pipeline needs to be on the fly, meaning it can be generated with lazy loading and streamed with the generator, to avoid out of memory (OOM) RAM crashes during training.

- **NFR3: Lightweight Footprint**

  To sustain the proposed network parameters are kept below 2.5 million during training, optimizing the inference latency and power consumption.

- **NFR4: Modular Architecture**

  The framework should be divided into different modules: self-supervised pretraining phase, supervised classifier training phase and joint fine tuning phase.

3.2 System Overview/ Architecture

The suggested system is designed in a two stages training pipeline. During Phase 1, the encoder is pre-trained to learn about the facial patterns and boundaries without labels. In Phase 2, the feature maps are clustered into groups with a Slot Attention module and the classifier is further fine tuned to recognize deepfakes.

3.2.1 High-Level Architecture

The proposed high-level architecture consists of two main stages: a self-supervised pretraining phase and a supervised fine-tuning phase. In Phase 1, a ResNet-like encoder extracts spatial feature maps, which are then passed to a Slot Attention module to group features into 8 object-centric representation slots. The encoder is pretrained using NT-Xent loss. In Phase 2, the slot representations are averaged and mapped to a binary classification head (comprising Dense, BatchNorm, ReLU, Dropout, and Sigmoid layers) to predict the probability of the image being real or fake.

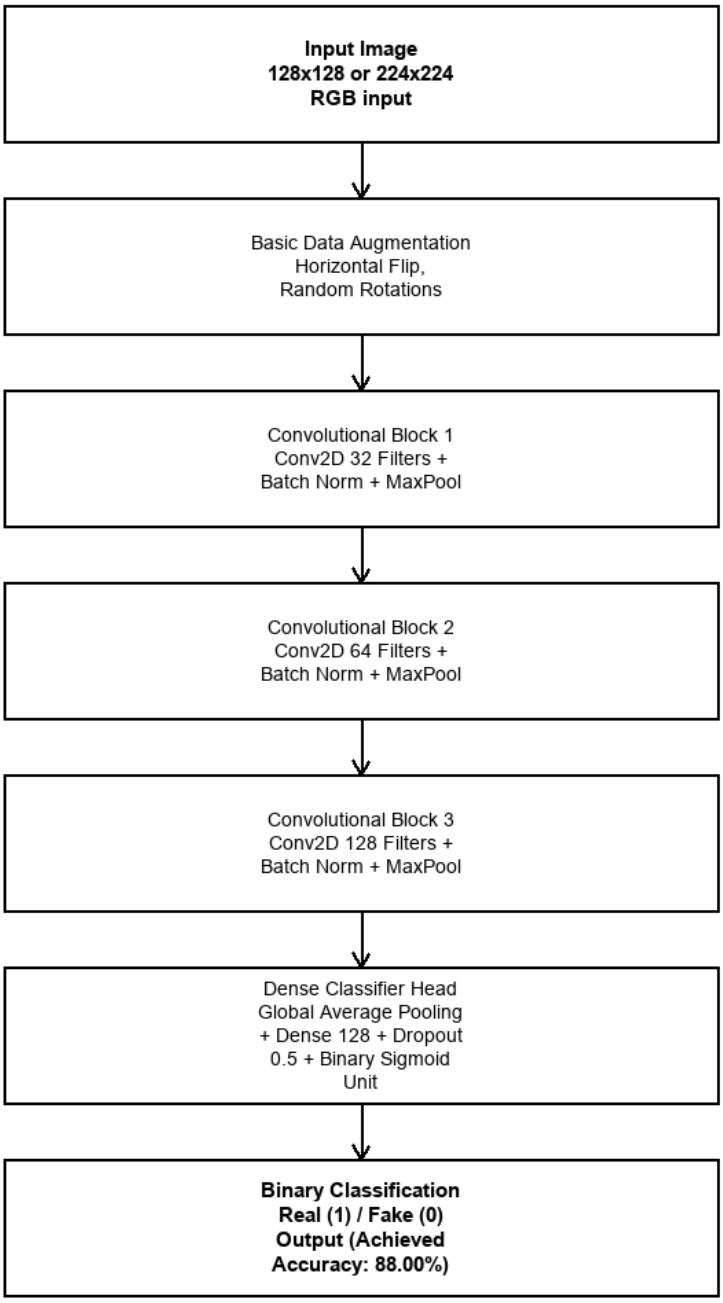


Figure 3.1: CNN Architecture (Phase 1)

Figure 3.1 illustrates the convolutional neural network (CNN) encoder architecture used in Phase 1 of our framework to extract robust, localized spatial features from input face images without relying on class labels.

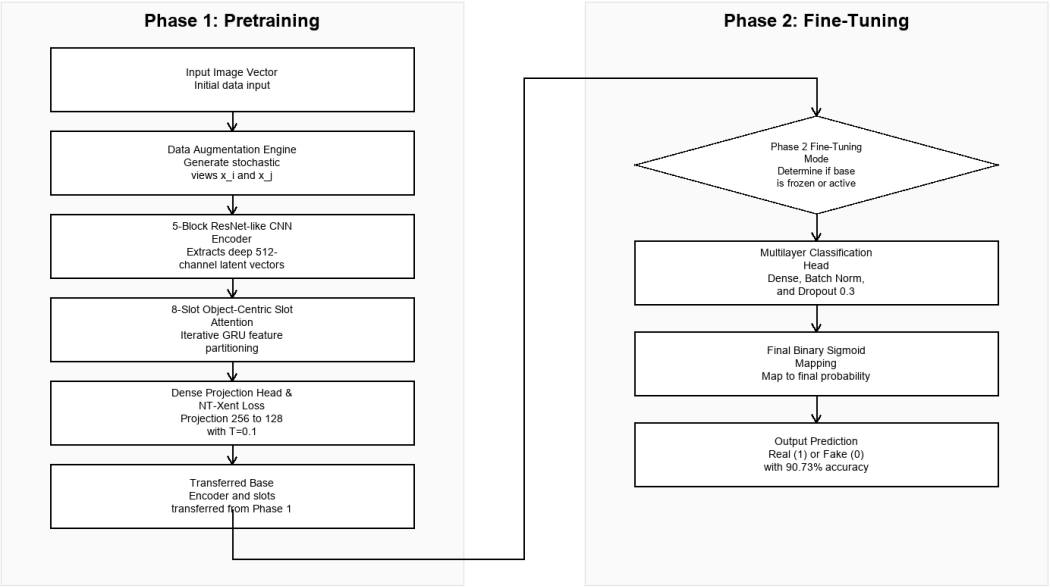


Figure 3.2: Supervised Learning Architecture (Phase 2)

Figure 3.2 depicts the supervised fine-tuning architecture in Phase 2, where the slot-based representations are mapped through a dense classification head to evaluate the probability of the image being forged.

3.2.2 Detailed System Diagrams

This section presents the detailed structural and behavioral diagrams of the proposed deepfake detection system.

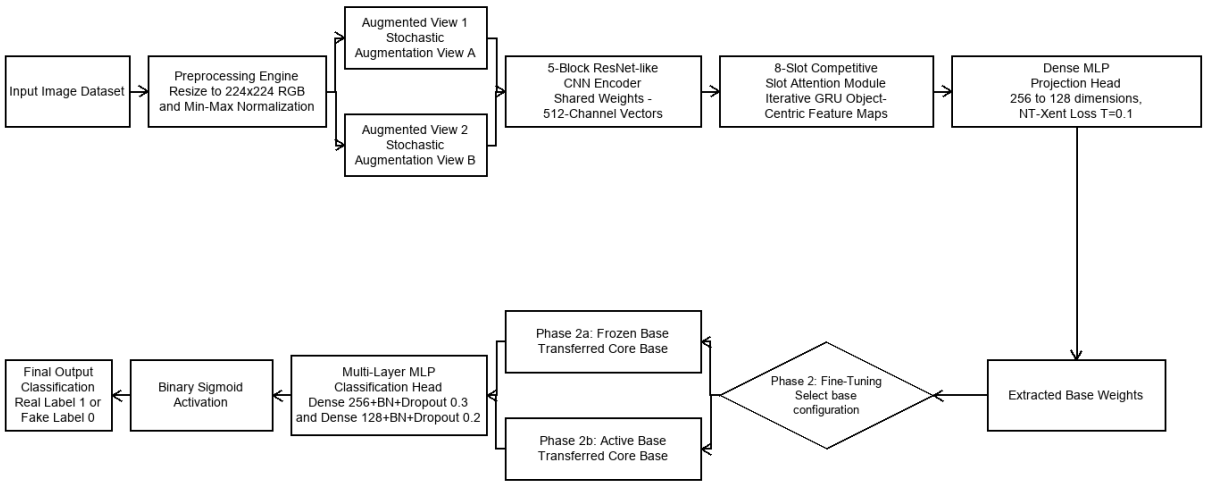


Figure 3.3: System Block Diagram

Figure 3.3 shows the modular block diagram of the system, illustrating the flow of data from the input pipelines to the self-supervised feature extraction and decision layers.

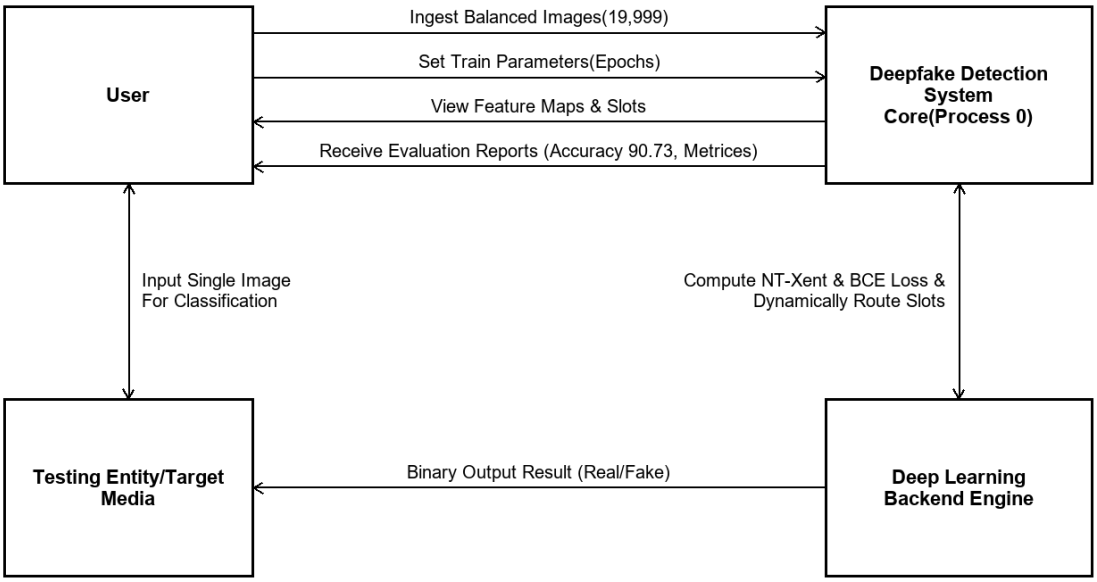


Figure 3.4: Data Flow Diagram (Level 0)

Figure 3.4 displays the Level 0 Data Flow Diagram (DFD), showcasing how image files are passed from external repositories to pre-processing, slot attention grouping, and metric evaluations.

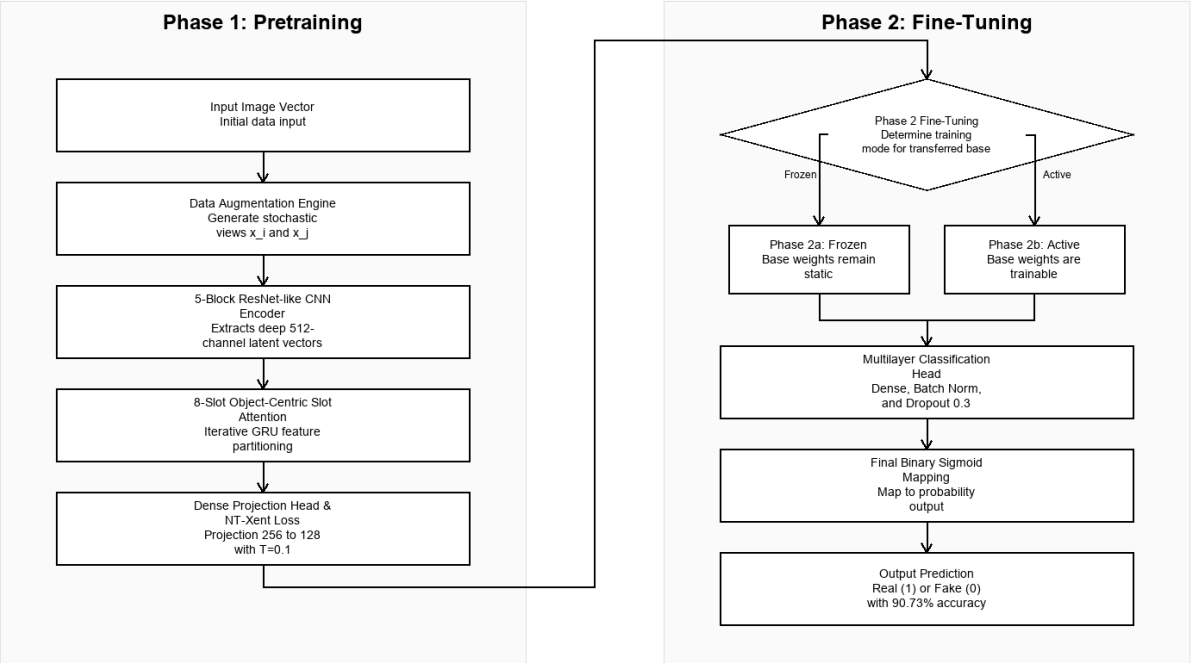


Figure 3.5: Flowchart of Complete Generation Pipeline

Figure 3.5 outlines the operational flowchart of the entire machine learning pipeline, tracking training states and early-stopping checks for both contrastive pretraining and downstream classification.

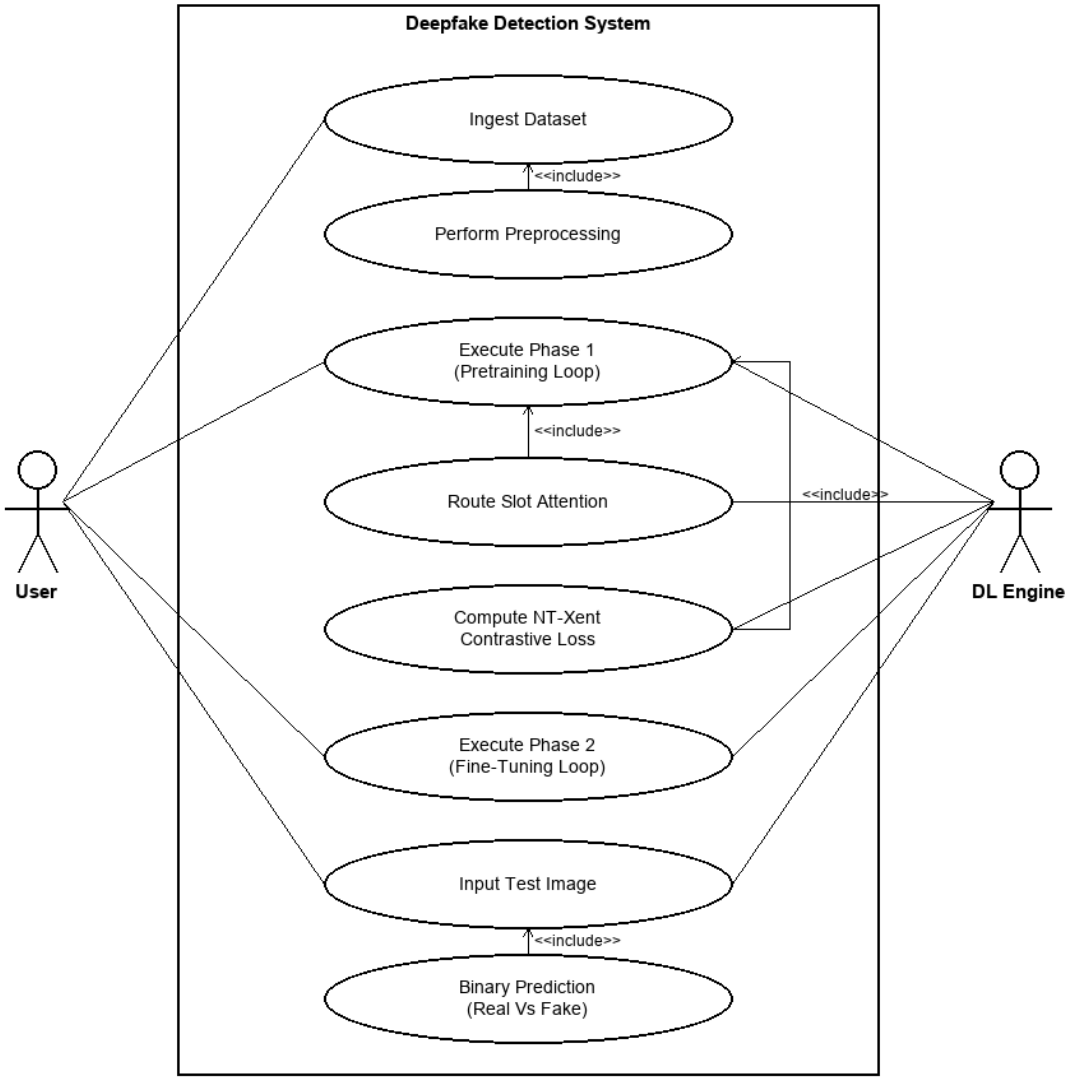


Figure 3.6: Use Case Diagram

Figure 3.6 represents the system's Use Case diagram, detailing the interactions of researchers and system administrators with the training modules, testing utilities, and visualization modules.

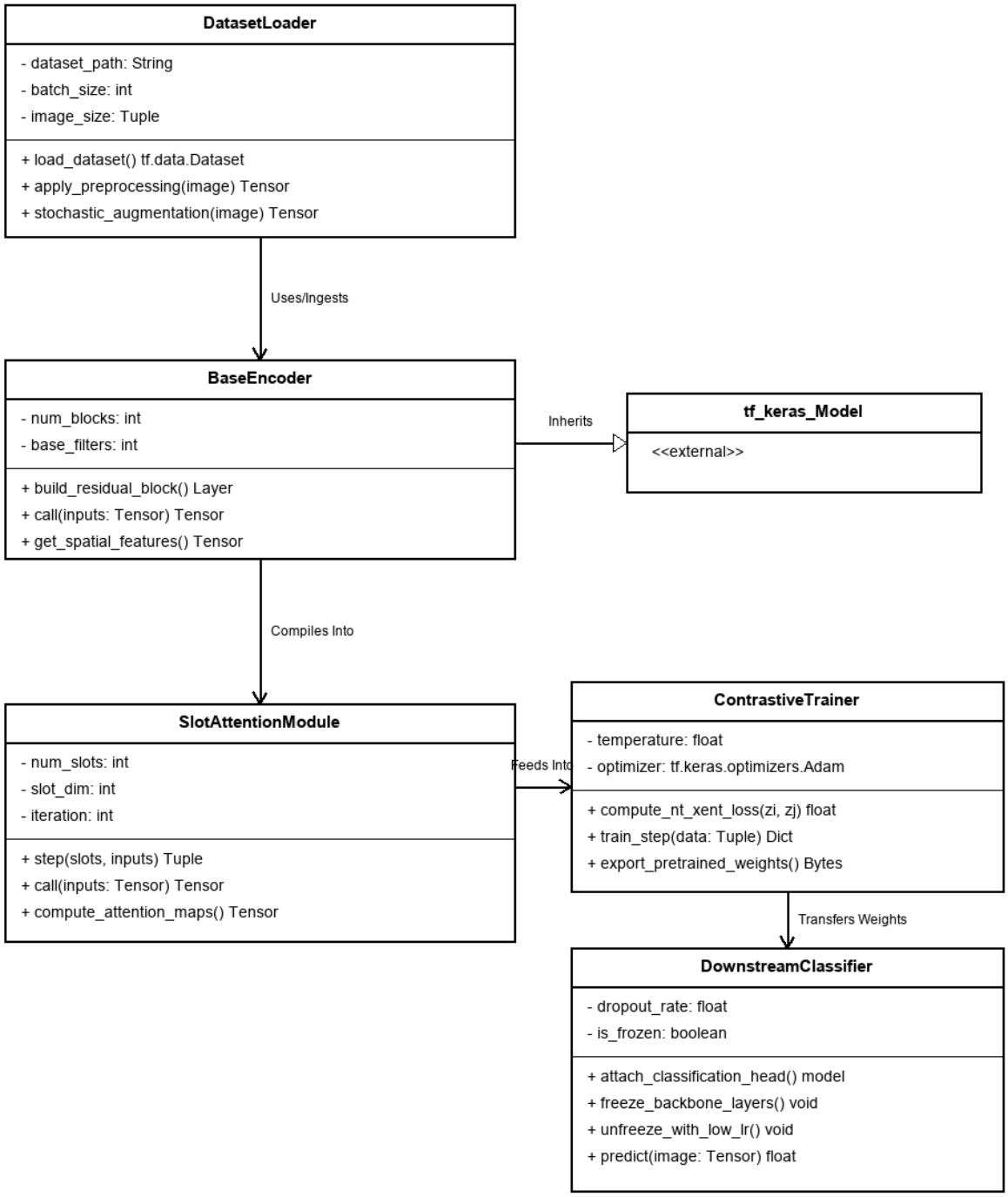


Figure 3.7: Class Diagram

Figure 3.7 details the static class structure of the implementation, documenting the attributes, methods, and inheritance links of the dataset managers and training frameworks.

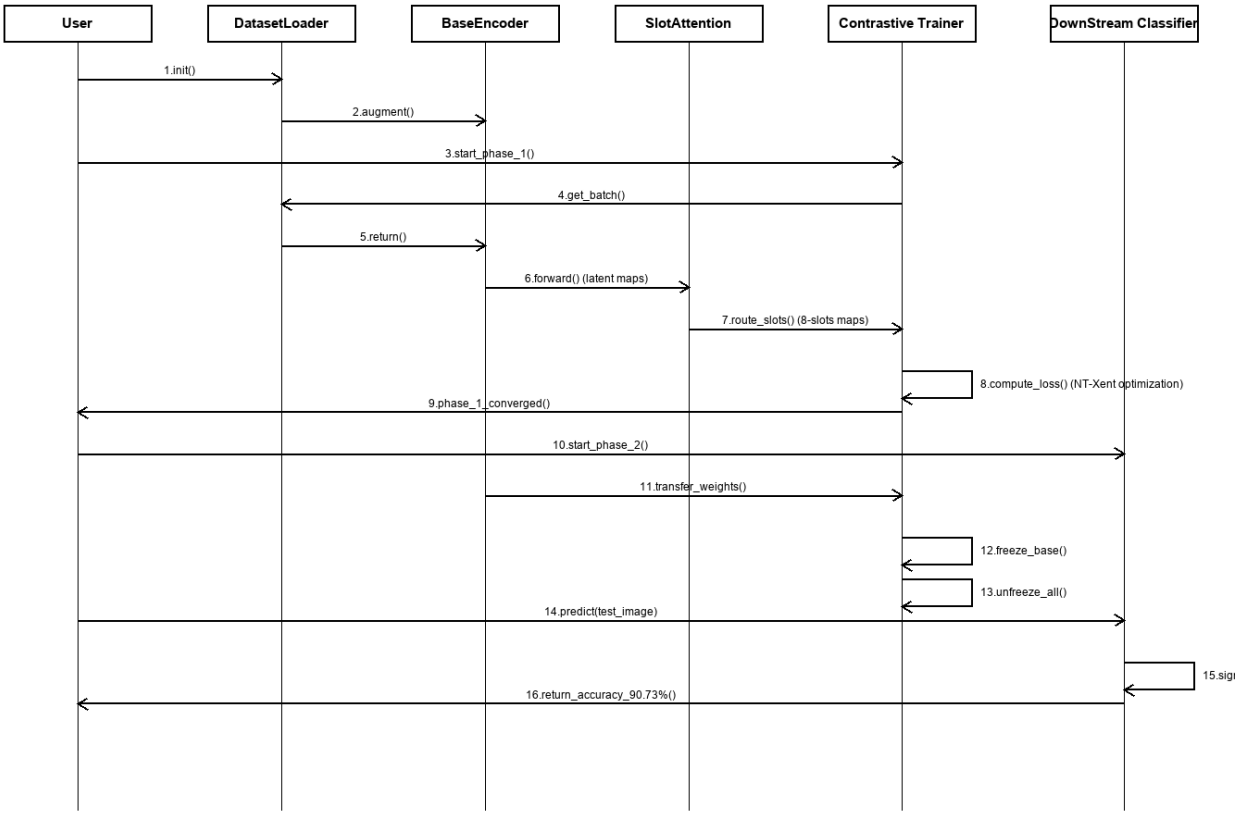


Figure 3.8: Sequence Diagram

Figure 3.8 describes the sequence of runtime actions and message passing between the data loader, neural layers, and optimization routines during a full epoch of training.

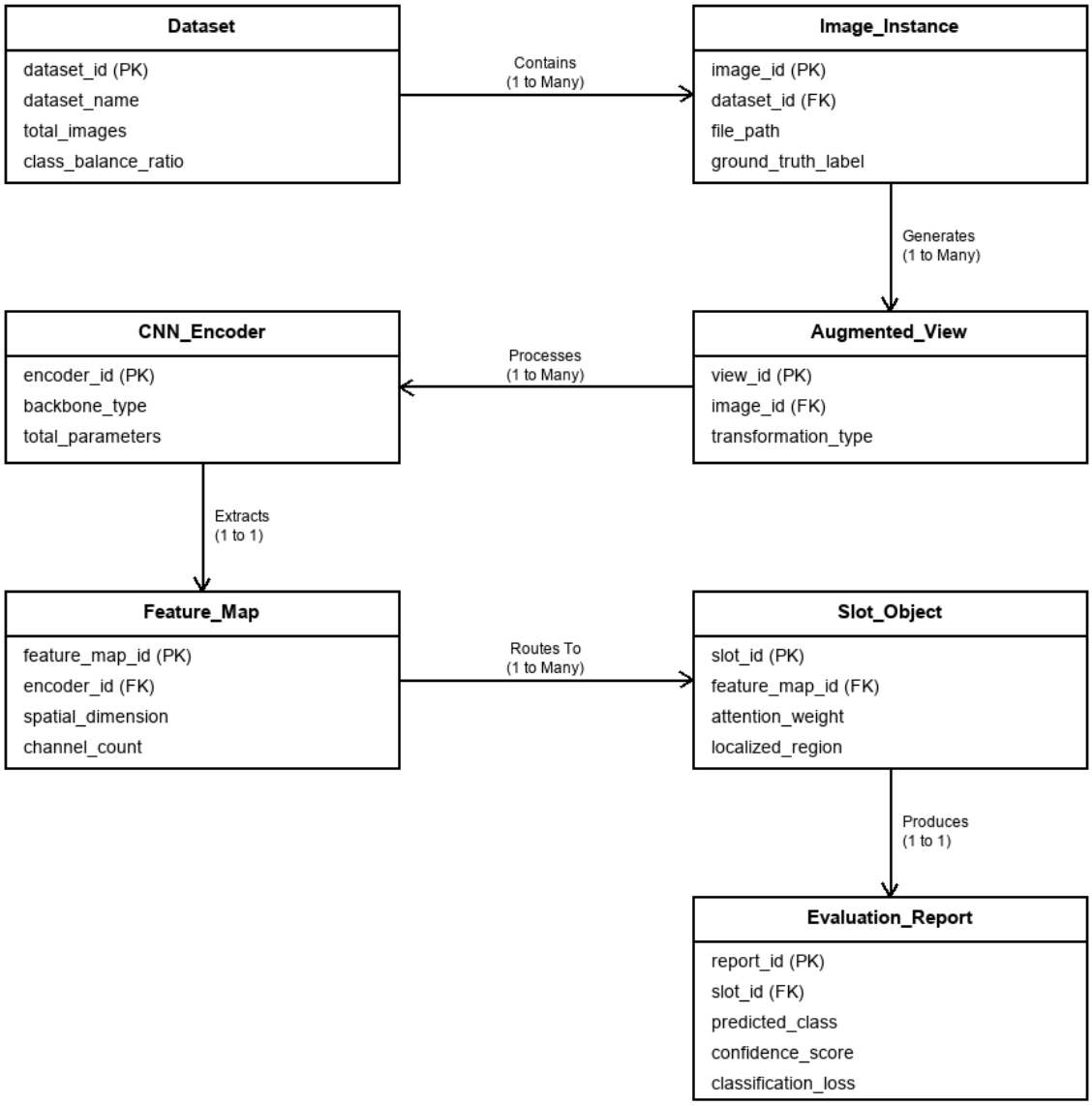


Figure 3.9: Entity-Relationship (ER) Diagram

Figure 3.9 displays the Entity-Relationship (ER) diagram of the metadata schemas, tracing the links between datasets, hyperparameter configurations, and epoch-wise performance logs.

Table 3.1 highlights the architectural alternatives evaluated before finalizing the system design:

Table 3.1: Architectural Design Decisions

Design Dimension	**Alternative Considered**	**Selected Design**	**Rationale for Selection**
Feature Extraction	Vision Transformers (ViT)	Custom ResNet like CNN	The ViTs demand huge amounts of data and have high memory demands on the GPU, leading to the crashing of RAM in limited environments. The ResNet like CNN has 5 residual blocks which gives a light weight footprint of ≈2.26M parameters.
Attention Module	Self-Attention (Transformer)	Slot Attention (Competitive)	Generic self-attention discovers global dependencies, which is overfitting to background textures. Slot Attention learns to normalize over slots, create competitive segmentation, while also isolating local artifacts.
Training Pipeline	End-to-End Direct Supervised	Contrastive Pretraining + Supervised Fine-Tuning	All of these are supervised models directly trained on shortcut features specific to the dataset. Pretraining teaches the network to learn generic facial textures first, thus better generalizing on deepfake models that it has never seen before.

Table 3.1 outlines the key architectural trade-offs guiding our design. By selecting a lightweight ResNet-like CNN, competitive Slot Attention, and a contrastive pretraining

strategy, the proposed system effectively balances computational efficiency, localization of facial features, and generalization to unseen deepfake variations.

3.3 Mathematical Model

The logic of the proposed system is modeled computationally as a series of mathematical operations.

3.3.1 Convolutional Feature Extraction

An input image $x \in \mathbb{R}^{H \times W \times 3}$ is mapped by the convolutional encoder $f(\cdot)$ to a feature map Y:

$$Y = f(x) \in \mathbb{R}^{H' \times W' \times D} \tag{3.1}$$

where H' and W' represent downsampled spatial dimensions and $D = 512$ is the channel dimension.

3.3.2 Competitive Slot Attention Matching

The spatial features are flattened to $Y \in \mathbb{R}^{HW \times D}$ and projected to Keys (K) and Values (V) using learnable matrices $W_k, W_v \in \mathbb{R}^{D \times D_v}$. The slots $S \in \mathbb{R}^{S \times D_v}$ are projected to Queries (Q) using $W_q \in \mathbb{R}^{D_v \times D_v}$:

$$K = YW_k \in \mathbb{R}^{HW \times D_v}, \quad V = YW_v \in \mathbb{R}^{HW \times D_v}, \quad Q = SW_q \in \mathbb{R}^{S \times D_v} \tag{3.2}$$

The competitive attention coefficients $A \in \mathbb{R}^{S \times HW}$ are computed using dot-product attention normalized over the slot dimension:

$$A_{s,i} = \frac{\exp\left(\frac{Q_s \cdot K_i^T}{\sqrt{D_v}}\right)}{\sum_{l=1}^{S} \exp\left(\frac{Q_l \cdot K_i^T}{\sqrt{D_v}}\right)} \tag{3.3}$$

The slot updates are calculated as the weighted sum of the values, and the slot representations are updated recurrently using a Gated Recurrent Unit (GRU):

$$U = A \cdot V \in \mathbb{R}^{S \times D_v} \tag{3.4}$$

$$S^{(t)} = \text{GRU}(S^{(t-1)}, U) \tag{3.5}$$

3.3.3 Classification Mapping

During Phase 2, the 8 slot representations $S \in \mathbb{R}^{8 \times 512}$ are averaged across the slot dimension to produce a global representation vector $v \in \mathbb{R}^{512}$. This vector is passed to the classifier head:

$$h_1 = \text{Dropout}(\text{BatchNorm}(\text{ReLU}(vW_1 + b_1)), p = 0.3) \tag{3.6}$$

$$h_2 = \text{Dropout}(\text{BatchNorm}(\text{ReLU}(h_1W_2 + b_2)), p = 0.2) \tag{3.7}$$

$$\hat{y} = \sigma(h_2W_3 + b_3) \tag{3.8}$$

where $\hat{y} \in [0, 1]$ is the output probability, and $\sigma(a) = \frac{1}{1+e^{-a}}$ is the sigmoid activation function.

3.4 Summary

Functional and non functional requirements for the deepfake detection system were set out, focusing on memory efficient pipeline executions. Custom CNN encoder with competitive 8 slot attention layer was chosen over demanding Vision Transformers or more globally focussed CNNs, for a lightweight, more generalizable alternative. Mathematical operations and training pseudocodes for the detection system were provided to complete the architectural design and system analysis section of this report.

Chapter 4

Methodology and Implementation

This chapter describes the methods and implementation of our proposed deepfake detection system. It outlines the overall methodological approach, identifies software and library dependencies and describes the OpenDeepfake Preview dataset. Subsequently, details of image preprocessing, network architecture design, modular implementation and hardware/-software environments are fully illustrated.

4.1 Methodological Framework

The development of the proposed system takes place primarily within an arranged methodological framework, as presented in a six phase sequential process on figure 4.1:

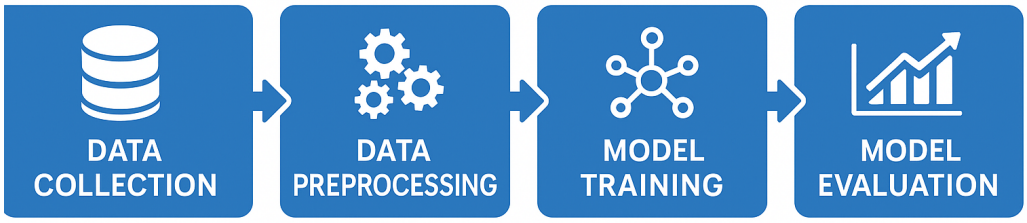


Figure 4.1: Proposed Methodological Framework

The methodology of the proposed work will consist of five phases. The first phase consists of loading of the dataset from Hugging Face and splitting into training, validation, and test datasets. The second phase includes the preprocessing step where the resizing, normalization, and augmentation are applied on the fly in order to make the data more diverse,

while augmented dual view images are generated for the contrastive pretraining. The third phase consists of the pretraining of the encoder with self-supervised learning approach and NT-Xent loss function for 100 epochs for the extraction of the facial representation features without using any labels. This is followed by the Phase 2a where the pre-trained encoder is frozen and only classification head is fine tuned for 12 epochs with the binary cross entropy loss function. The final phase, Phase 2b includes the fine tuning of both encoder and classification head for 50 epochs at reduced learning rate for the adaptation of learned feature representations for binary classification.

4.2 Tools, Libraries and Technologies used

The proposed architecture will be implemented with the help of a state of the art deep learning environment that includes the following languages and libraries: Python (version 3.10 or higher) will be the main programming language for coding, developing models and analyzing data; TensorFlow and Keras (version 2.15 or higher) will be used to implement the convolutional blocks, Slot Attention layers, self-supervised pre-training loop, and classification head; TensorFlow (version 2.15 or higher) will also provide the needed functionalities to load and process the Hugging Face datasets; Hugging Face Datasets will be used for streaming and caching of the OpenDeepfake Preview dataset; OpenCV (version 4.8 or higher) will be used for image processing and transformation of the PIL images into suitable formats for real-time input into the model; NumPy and Scikit learn will be used for matrix operations, splitting of datasets and calculation of evaluation metrics, such as Precision, Recall, F1-Score, and ROC-AUC; Matplotlib will be used for visualizations of training and validation curves, Slot Attention output, ROC curve, etc.

4.3 Data Collection/ Data Description

The model was put in its paces with the help of the OpenDeepfake-Preview Dataset (Hugging face made).

4.3.1 Dataset Statistics

The dataset is about as balanced as you can get. Out of 19,999 images, 9,999 are real (labeled 1, making up 49.998%) and 10,000 are fake or manipulated (labeled 0, which comes out to 50.002%). So, you won't run into any problems with the model favoring one class over the other.

4.3.2 Splitting of the Datasets

In order to measure the ability of generalization of the proposed model, a split of the dataset into training, validation, and test datasets is carried out via stratified sampling. In the training set, which accounts for 70% of the dataset, 13,999 images (7,000 fake and 6,999 real) are used for model learning. The validation set, which is 15% of the dataset and contains 3,000 balanced images (1,500 fake and 1,500 real) serves as the data for tuning the parameters of the model and observing the learning process in order to find the optimal point of stopping in order to avoid overfitting. The rest of the dataset, 15%, which also includes 3,000 balanced images, serves as the test set.

4.4 Preprocessing Steps

In order to maintain consistency in terms of the quality of the inputs as well as prevent overfitting, three types of preprocessing are done to the images before they are utilized in the proposed model. Firstly, the images are resized in order to ensure that they are all of equal resolutions and thus all have an equal input size. Secondly, the images are normalized through the use of min-max normalization in order to ensure that they all lie within the same pixel range.

4.4.1 Resolution Standardization

Firstly, all input images are standardized with regard to their image resolution by converting them to the RGB format. Due to hardware limitations, two resolutions are used as inputs

to models based on the computing platform. In the local environment, with an NVIDIA GeForce MX350 GPU, input images are down sampled to 128×128. In the cloud environment, like Kaggle and Google Colab, input images are downscaled to 224×224 in order to retain fine manipulations and blending discrepancies.

4.4.2 Min-Max Normalization

The pixel values are normalized using the method of min max normalization in which the pixel values ranging from $[0, 255]$ are mapped to a new range $[0.0, 1.0]$:

$$I_{norm} = \frac{I_{raw}}{255.0} \tag{4.1}$$

This normalization improves numerical stability during training and ensures consistency across input samples.

4.4.3 Stochastic Data Augmentation

This is accomplished by double passing the training images via the stochastic augmentation pipeline that is repeated twice before pre-training stage in the first phase in order to obtain two different perspectives of the same image semantically. It is accomplished using different types of augmentations including random horizontal flip, random rotation at 8% of the whole rotation (which is equal to about 28.8 degrees), random contrast variation by 15%, and random brightness variation by 15%.

4.5 Model Development and Algorithm Design

There are three main parts in the model: convolutional encoder, Slot Attention layer, and the classification part. These layers are responsible for the extraction of discriminative face features and the final real/fake classification task, respectively.

4.5.1 Convolutional Encoder Architecture

The model of the convolutional encoder implements ResNet for extraction of the hierarchy of spatial features from the input image. There are five consecutive convolutional layers that make up the structure of the convolutional encoder model. In the first place, an input image (H x W x 3) passes through Block 1 with 32 filters of (3 x 3) convolution, batch normalization, ReLU activation, and max pooling. The same pattern goes for Blocks 2, 3, and 4 with 64, 128, and 256 filters respectively, (3 x 3) convolution, batch normalization, ReLU activation, and max pooling. Lastly, the Block 5 is made up of 512 filters with (3 x 3) convolution and ReLU activation with batch normalization. Hence, the hierarchy of feature maps (H/16 x W/16 x 512) is extracted from the input image and passed on to the next Slot Attention layer. Skip connections in all residual blocks make it possible to maintain data from previous layers, at the same time making training easy. Residual function is formulated as follows:

$$Y_{block} = \text{ReLU}(\text{BN}(\text{Conv}(X)) + W_s(X)) \tag{4.2}$$

Where X represents the input to the block, and W_s denotes 1×1 convolution to ensure that the dimensions of features are the same in case downsampling is required.

4.5.2 Slot Attention Module

Feature maps obtained from the encoder network are converted into a sequence of spatial tokens, which then becomes an input to the Slot Attention network. In the case of a 224×224 input size, the feature map has a size of $14 \times 14 \times 512$, and in the case of a 128×128 input size, the feature map has a size of $8 \times 8 \times 512$. The above feature maps are flattened out as follows:

$$Y \in \mathbb{R}^{HW \times 512} \tag{4.3}$$

The Slot Attention module is made up of 8 slots, each 512 dimensioned. Values are sampled from a common Gaussian distribution and refined in three steps in an iterative fashion with dot product attention mechanisms. In each step, slots compete for the useful spatial features in the encoder output space, thus creating latent groups in the output space of the encoder. The above process allows the model to attend to the specific regions of the face, for instance,

eyes, mouth region, or skin color irrespective of the rest of the components.

4.5.3 Classification Head

The classification head uses the slot representations to predict the probability of the image belonging to the real class using binary classification. In the first place, the mean value of the eight slot vectors produces a single (1×512) feature vector that serves as an input to the classifier. The classifier has a fully-connected layer with 256 neurons using the ReLU activation function, along with batch normalization and a dropout of 30%. Further to it is another fully-connected layer with 128 neurons and the ReLU activation function, batch normalization, and a dropout of 20%. Finally, there is only one neuron with sigmoid activation that predicts the probability of the image being real.

4.6 System Implementation and Module Description

There are three modules used in the implementation of the proposed approach, namely: memory friendly image data generation, self-supervised pre-training, and classification model fine tuning.

4.6.1 Memory-Efficient Data Generator Module

For the avoidance of loading all 19,999 images in the memory, the custom data generator is used for the real time image decoding and preprocessing. The generator processes every image by converting it to the RGB format, changing its size to match the input size and normalizing the pixel values to use the processed image during the training. As a result, only a batch of images is loaded into the memory at once, and this allows training the model even when the VRAM of the system is limited. The generator is incorporated within the `tf.data.Dataset` pipeline.

4.6.2 Phase 1: Pretraining Module

The encoder is trained initially through Cosine Decay of the Learning Rate. In this case, the value of the learning rate is made 3×10^{-4} and then decreased to 3×10^{-5}. The network is then trained by making use of Adam optimizer and NT-Xent Loss with $\tau = 0.1$. The unsupervised learning model is then executed for 100 epochs. This will enable training of compact representations of faces from the images without class labels.

4.6.3 Phase 2: Classification Fine-Tuning Module

The second stage of the training process entails two stages. The second step in the training process will be: The Encoder will be frozen for 12 epochs with a learning rate of 1×10^{-4} and the classification head will be trained for 100 epochs with a learning rate of 1×10^{-4}. The second phase of the learning process will be repeated with the learning rate of 5×10^{-5} and without the encoder being frozen, for 50 epochs. The weight decay is equal to 1×10^{-4} so that there will be generalization of results. Early stopping will be done to avoid over fitting.

4.7 Hardware and Software Requirements

Table 4.1 lists the hardware and software configurations used to develop and evaluate the system.

Table 4.1: Implementation Environments

Configuration Item	**Local Environment**	**Cloud Environment (Kaggle/-Colab)**
Operating System	Windows 11 (64-bit)	Linux Ubuntu 22.04 LTS
CPU	Intel Core i7-10750H @ 2.60GHz	Intel Xeon @ 2.20GHz
System Memory (RAM)	16 GB DDR4	16 GB DDR4
GPU / VRAM	NVIDIA GeForce MX350 (2 GB VRAM)	NVIDIA Tesla T4 (16 GB VRAM)
Image Resolution	$128 \times 128 \times 3$ pixels	$224 \times 224 \times 3$ pixels
Batch Size	16	64
Python Version	3.10.12	3.10.12
DL Framework	TensorFlow 2.15.0	TensorFlow 2.15.0

Table 4.1 outlines the dual-setup approach used for our implementation. The local configuration provides a lightweight debugging platform with smaller batch sizes and resolution, while the cloud setup leverages higher VRAM to execute the full-scale model training and pretraining phases.

4.8 Summary

In this chapter, the technique and deployment of deepfake detection will be discussed. It provided the technique and presented the balance.OpenDeepfake Preview dataset Image pre-processing process and architecture of the model was described. At last, its memory efficient dataset generator and system requirements will be discussed.

Chapter 5

Results and Evaluation

This chapter focuses on the analysis and performance evaluation of the system that is designed for deepfake detection. Description of the experimental setting is given, metrics of performance are stated, and comparison with the CNN baseline system is carried out. At last, important results are discussed, and the drawbacks of the current method are highlighted.

5.1 Experimental Setup

To test scalability of the model in relation to the hardware availability, the experiments were carried out both locally and in the cloud. Our model implementation was done with Python 3.10 in combination with TensorFlow and Keras frameworks.

For the evaluation purpose, we used 3,000 pictures from the balanced OpenDeepfake Preview dataset that consisted of 1,500 real and 1,500 fake pictures. Hyperparameters used during the training of the model and the baseline are described in Table 5.1.

Table 5.1: Training Hyperparameter Configurations

Hyperparameter	**Proposed Framework (Pretraining)**	**Proposed Framework (Fine-tuning)**	**Baseline Supervised CNN**
Optimizer	Adam	Adam	Adam
Initial Learning Rate	3×10^{-4} (Cosine Decay)	5×10^{-5}	1×10^{-4}
Batch Size	64	64	64
Epochs	100	50 (Early Stopping)	30 (Early Stopping)
Loss Function	NT-Xent ($\tau = 0.1$)	Binary Cross-Entropy	Binary Cross-Entropy
Input Dimensions	$224 \times 224 \times 3$	$224 \times 224 \times 3$	$224 \times 224 \times 3$

Table 5.1 outlines the training configurations across different models, highlighting that while batch sizes and input dimensions remain consistent, the proposed framework utilizes NT-Xent loss with cosine decay during unsupervised pretraining before switching to binary cross-entropy and a lower learning rate for fine-tuning.

5.2 Evaluation Metrics

We assess our classification performance using five standard metrics:

Accuracy calculates the proportion of correct predictions, including both real and fake images, out of the total test set:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.1}$$

Here, TP represents True Positives, TN is True Negatives, FP is False Positives, and FN is False Negatives.

Precision shows how many of the predicted positive cases are actually positive. It is the ratio of true positives to all positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.2}$$

Recall (or sensitivity) measures how well the model identifies actual positive cases. It is the ratio of correctly predicted positives to the total number of real positives:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5.3}$$

The F1-Score is the harmonic mean of precision and recall. It gives us a balanced measure of both metrics:

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5.4}$$

The Area Under the ROC Curve (ROC-AUC) measures how well the model separates real and fake images across all classification thresholds. We plot this by comparing the True Positive Rate (TPR) against the False Positive Rate (FPR):

$$\text{TPR} = \frac{TP}{TP + FN} \tag{5.5}$$

$$\text{FPR} = \frac{FP}{TN + FP} \tag{5.6}$$

5.3 Result Analysis

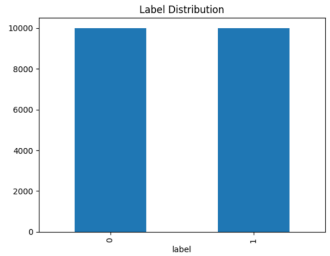
The framework presented was tested on the test data after the entire training procedure was completed. The performance metrics are provided in Table 5.2 below.

Table 5.2: Classification report for our model

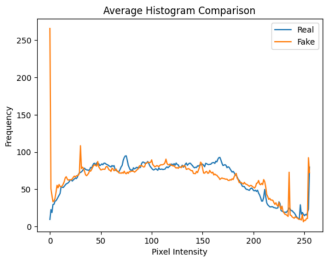
Class	**Label**	**Precision**	**Recall**	**F1-Score**	**Test Accuracy**
Fake Image	0.0	0.94	0.87	0.90	-
Real Image	1.0	0.88	0.94	0.91	-
Overall Model	-	-	-	-	90.73%

Table 5.2 presents the classification performance of our model, showing an overall test

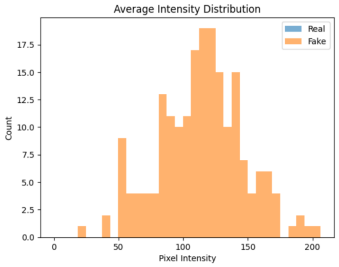
accuracy of 90.73%. Notably, the model achieves a high precision of 0.94 in identifying fake images, while maintaining a robust recall of 0.94 for real images, indicating its effectiveness in minimizing false alarms.



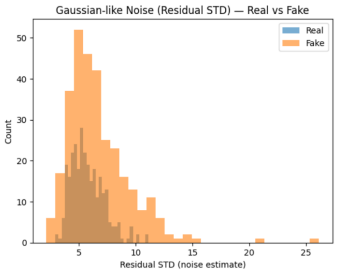
(a) Dataset Class Distribution: Shows the balanced class distribution of the dataset, consisting of exactly 9,999 real and 10,000 fake images.



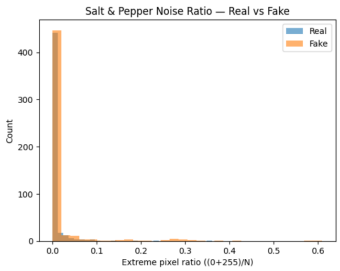
(b) Average Pixel Intensity Histogram: Compares the frequency distribution of overall pixel values between real and fake images.



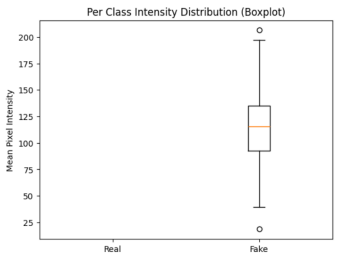
(c) Intensity Distribution Histogram: Displays the density and overall spread of average pixel intensities across all dataset samples.



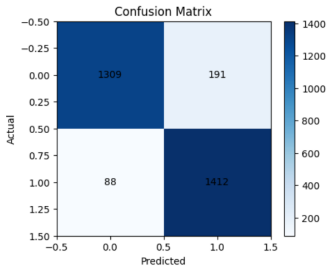
(d) Gaussian Noise Residual STD: Illustrates the estimated standard deviation of Gaussian noise residuals in real vs. fake images.



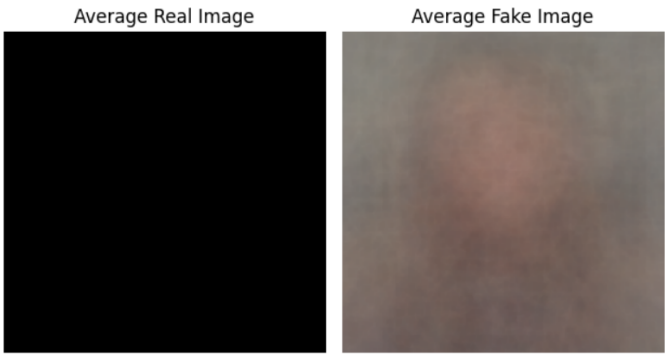
(e) Salt & Pepper Noise Ratio: Highlights the presence and density of impulse (salt and pepper) noise artifacts in the image classes.



(f) Per-Class Intensity Boxplot: Details the statistical distribution (median, quartiles) of average pixel intensities per image class.



(g) Confusion Matrix of Baseline Model: Presents the classification performance of the baseline CNN model, showing true and false predictions.

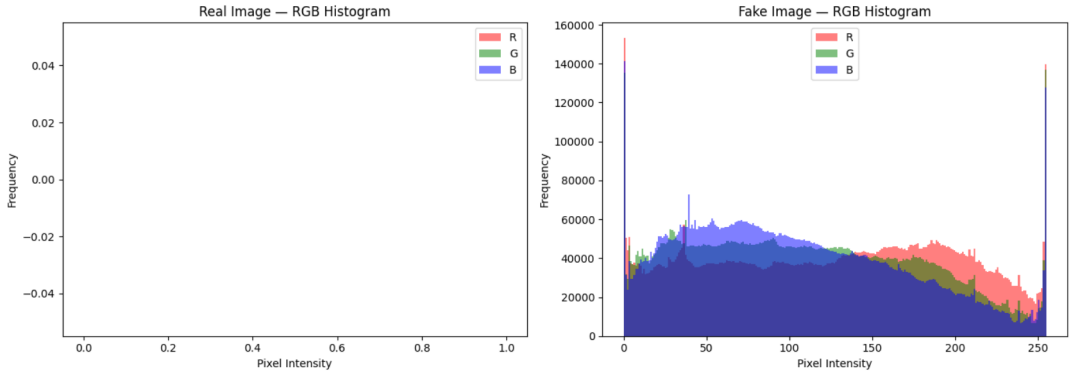


(h) Average Face Visualizations: Visualizes the average composite face of real versus fake images to highlight global structural trends.

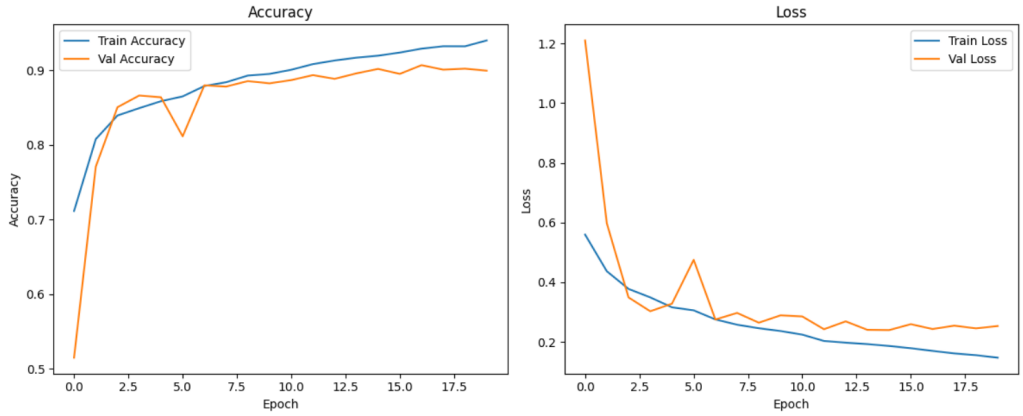
Figure 5.1: Baseline CNN Model Training and Evaluation Metrics (Part 1)

Figure 5.1 provides a comprehensive statistical analysis of the dataset alongside baseline

model predictions. While global pixel intensity metrics overlap significantly between real and fake images, microscopic noise residuals and boundary blending artifacts serve as distinct indicators of manipulation. Additionally, the confusion matrix shows that the baseline model successfully identifies the majority of samples, though with a notable rate of false alarms.



(a) RGB Color Space Histograms: Displays the color channel (Red, Green, Blue) frequency distributions for real and fake image classes.



(b) Training Curves of Baseline CNN: Tracks the training and validation accuracy and loss over 20 epochs for the baseline CNN model.

Figure 5.2: Baseline CNN Model Training and Evaluation Metrics (Part 2)

Figure 5.2 details the baseline CNN's training progress. The color histograms indicate highly similar distributions across channels for both classes, while the training curves reveal a clear performance bottleneck, with the validation loss beginning to plateau and fluctuate past the eighth epoch due to overfitting.

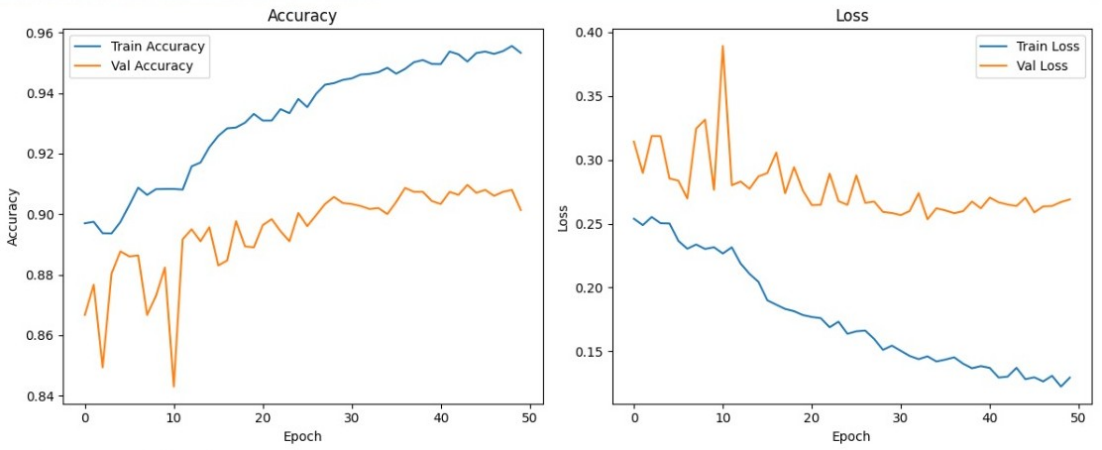


Figure 5.3: Self-Supervised Model Training and Evaluation Metrics: Tracks the training and validation accuracy and loss over 50 epochs for the proposed self-supervised model.

Figure 5.3 demonstrates the superior training stability of the proposed self-supervised framework. By utilizing contrastive pretraining followed by fine-tuning, the model achieves a higher, stable accuracy of 90.73% without showing typical signs of early overfitting, confirming the robustness of the learned features.

5.3.1 Key Observations

It can be seen from the experimental findings that the proposed approach has the capability of distinguishing between forged and genuine images. As regards the fake image category, the model has achieved precision of 0.94 meaning that 94% of the images that have been identified as fake belong to this category. Moreover, the recall score of 0.87 means that 87% of all fake images within the test data set have been identified using the model. The precision of 0.94 has also been obtained for the real image category, which implies that the model can properly authenticate the genuine images while making minimal mistakes. Generally, the proposed approach is 90.73% accurate when it comes to deepfake image identification.

5.4 Comparison with Baseline / Existing Approaches

To assess the improvements, the proposed framework was compared against a traditional fully supervised CNN baseline trained under identical dataset conditions. The comparative

results are summarized in Table 5.3.

Table 5.3: Performance Comparison

Architecture Model	**Parameters**	**Test Accuracy**	**Fake F1-Score**	**Real F1-Score**
Baseline Supervised CNN	$\approx 140,000$	88.00%	0.87	0.88
Proposed Framework	$\approx 2,260,000$	90.73%	0.90	0.91

The proposed architecture exhibits a comparative analysis of how it outperforms the baseline model on various fronts. The proposed architecture manages to achieve an accuracy level of 90.73%, indicating a rise of 2.73% when compared to the baseline model, where the accuracy level was 88.00%. In addition, although the proposed model has a smaller number of parameters (approximately 140k), it starts to exhibit signs of overfitting from the 8th training epoch. This indicates that the model will start learning the noise in the dataset rather than the features associated with the forgery. But the proposed architecture has excellent generalization capabilities.

5.5 Discussion on Findings

The empirical findings highlight a number of key advantages of the proposed approach. First, the use of self-supervised contrastive pre-training based on the NT-Xent loss allows the encoder to obtain reliable structural and boundary representations of facial features. Through maximization of the similarity between various views of the same image, the encoder becomes more robust to changes in brightness and color and focuses on relevant structural features. Second, the Slot Attention technique is able to filter out the background information through separation of the feature map into eight object centric slots through a competitive softmax technique. In contrast to the traditional CNN based approaches which are dependent on global pooling of the feature maps, the suggested architecture makes it possible to separate the crucial facial elements such as eyes, nose, and mouth boundaries from the static background information, thus minimizing the overfitting to the background.

Finally, the suggested approach increases robustness and interpretability through prediction based on localized slot representations in contrast to global embedding of feature maps. This allows the network to concentrate on localized blending artifacts characteristic of deepfake images, while attention maps serve as the explanation of the regions used for making predictions.

5.6 Limitations

However, the approach under discussion is accompanied by some drawbacks as well. Training procedure becomes quite complicated as there are three different steps involved self-supervised contrastive pre-training, training of the classification head and fine tuning of both encoder and classification head as one process. Such training procedure requires more computational power than the conventional one-stage training based on supervision. In addition, the described framework is currently designed for the analysis of static images and does not involve any temporal modeling. That is why it is impossible for the proposed technique to utilize temporal artifacts like unnatural blinking and jitter in order to detect deepfake videos. Furthermore, the model depends on the resolution of images as the fine details might be lost during the resizing to (128×128) pixels.

Chapter 6

Project Planning and Budget

In this chapter, the project timeline and its financing plan have been explained.

6.1 Project Timeline

To organize the development process better and ensure the adherence to schedule, the project has been divided into five stages totalling a whole year from July 2025 to June 2026. Project Timeline has been prepared with the help of the GanttProject tool. Schedule of the activities of the project stages is presented in Table 6.1.

Table 6.1: Project Gantt Chart Showing Timeline of Major Phases (July 2025 - June 2026)

Project Phase	**Jul '25**	**Aug '25**	**Sep '25**	**Oct '25**	**Nov '25**	**Dec '25**	**Jan '26**	**Feb '26**	**Mar '26**	**Apr '26**	**May '26**	**Jun '26**
Requirement Analysis	■	■	■									
System Design				■	■	■						
Implementation							■	■	■			
Testing & Validation										■	■	
Documentation & Report											■	■

Table 6.1 illustrates the chronological timeline of the project, detailing how the 12-month duration is structured into distinct, sequential phases from July 2025 to June 2026 to ensure systematic progression from analysis and design to implementation, testing, and final documentation.

Project Phases Description:

- **Requirement Analysis (Jul '25 - Sep '25):** Literature search on the newest existing deepfake detection frameworks, analysis of the size and class balance of the Hugging Face OpenDeepfake Preview dataset and the formulation of functional and non functional requirements.
- **System Design (Oct '25 - Dec '25):** System design of ResNet impacted the design of the encoder system, Slot Attention competitive mapping layers and classification heads. Mathematical modeling of the mathematical model of the GRU, mathematical modeler update is the preparation for NT-Xent loss.
- **Implementation (Jan '26 - Mar '26):** Efficient memory generator implementation on the fly batch loading. It was done to be able to run the self-supervised pretraining loop (Phase 1) and classification heads training (Phase 2) on the rented cloud Tesla T4 GPUs. Usage of these for training was as follows: Phase 1: self supervised pretraining loop; Phase 2: classification heads training in the rented cloud Tesla T4 GPU instances.
- **Testing and Validation(Apr '26 - May '26):** Hyperparameters tuning, validation of weights, configuration of early stopping, generating slot attention maps, calculation of final test accuracy (90.73%).
- **Documentation and Final Report(May '26 - Jun '26):** Preparation of chapters, writing of final thesis, presentation for the thesis defense and submission of the final thesis.

Note: No deviations from the schedule due to using cloud based approach that prevented any delays in local compilation process.

6.2 Budget Estimation

The estimation of the cost and the actual cost of the project are discussed in this part. A budget has been drawn up in order to show how economically feasible the project and how budget friendly it is.

Because of the complications associated with the training of the model through the process of contrastive pretraining on 19,999 images, the local machine with an MX350 GPU was inadequate in order to conduct training with the use of the large batch size. Hence, the cloud GPU NVIDIA Tesla T4 (16 GB VRAM) was rented.

Table 6.2: Estimated Project Budget

Item / Resource	**Description / Source**	**Quantity**	**Unit Cost (BDT)**	**Total Cost (BDT)**
Cloud GPU Rental	NVIDIA Tesla T4 (16 GB VRAM)	4 Months	3,000	12,000
Internet & Utility Fees	High-speed connection for dataset caching	4 Months	1,000	4,000
Software Tools / APIs	Python, TensorFlow, HF	-	Open Source	0
Miscellaneous	Thesis book printing, binding, transport	-	1,500	1,500
Total Estimated Budget				**17,500 BDT**

Table 6.2 outlines the estimated project budget, showing a total expenditure of 17,500 BDT. The primary driver of this cost is the cloud GPU rental (12,000 BDT) needed for high-performance pretraining, supplemented by internet utility fees and minor miscellaneous expenses for thesis finalization.

6.3 Summary

This chapter gives an overview of the schedule and funding for the deepfake detection research. The research has been done systematically in 12 months, from July 2025 to June 2026. There were five stages included in the process, which were done systematically through the use of GanttProject software. In the allotted period, all the milestones related to data preparation, self-supervised contrastive pretraining, and baseline testing were completed by the researchers.

From the perspective of finance, the cost management plan includes making an investment in high end cloud computing facilities so as to overcome the limitations related to hardware capacity without which training has not been possible on the dataset of 19,999 images. With the help of renting a NVIDIA Tesla T4 GPU, all activities of the training have been conducted in an efficient way in order to keep the cost under control. Taking into account

all costs associated with cloud computing, data usage, and hard copy, the total cost of the project has been kept below the budget of 17,500 BDT.

Chapter 7

Standards and Design Constraints

An example of how to satisfy the prerequisites and accreditation for CEP and CEA is addressed in this chapter. Achieved adherence to professional standards and evidence for Knowledge Profiles, CEP & CEA.

7.1 Compliance with standards and professional practice

It is very important that all the aspects of deep learning technology should comply with well established codes and standards of engineering practices. In this section, the compliance with the standards is discussed along with the inclusion of societal, environmental and safety aspects in the choice of hardware and software components.

7.1.1 Software Standard

The software development of the proposed software is based upon several well-known software engineering standards to ensure software quality, security, maintainability and reliability. The software development in general follows the ISO/IEC 12207 standard, called Software Life Cycle Processes. It proposes a systematic SDLC, which involves requirement analysis, system architecture design, modular implementation and software validation. To guarantee data security, the system is based on the principles of ISO/IEC 27001 standard under Information Security Management, because the deep fake detection system operates on the biometric information in the form of a facial image. All the pre-processing and

inferential activities are carried out in secured local and cloud environment and the user images used for inference are not saved permanently following the GDPR principles. In addition, it complies with a standard style guide for Python (PEP 8). It ensures uniformity of codes and better readability and maintainability of the source code. Finally, the models developed and evaluated follow the principles of IEEE 1012-2016 standard for System, Software and Hardware Verification and Validation using separate datasets for training, validation and testing from OpenDeepfake Preview dataset.

7.1.2 Hardware Standard

The efficiency of deep learning models depends highly on the computer infrastructure driving the models. As a result, the computer infrastructure in the training and inference process in deep learning models is compliant with the standards in computing. The computer infrastructure complies with the standards set by NVIDIA CUDA for efficient parallel processing, and efficient floating-point computations on NVIDIA GPUs, including the local NVIDIA MX350 GPU and NVIDIA Tesla T4 GPU on cloud computing infrastructure. Furthermore, the computer infrastructure complies with the standards set out by IEEE 1680 Environmental assessment of personal computer products, in terms of efficient designs of computers and servers in order to reduce energy consumption in computation and minimizing the environmental impact.

7.1.3 Others

The model is based on the components related to the ethics and risks management of the artificial intelligence technology. The process of development of the system occurs according to the IEEE 7000-2021 standard. In this paper, there is proposed a framework that is based on the ethics of development of the system and can help to solve such important issues as privacy, transparency, accountability, and prevention of any misuse of the algorithms of artificial intelligence. Besides, the proposed framework satisfies the requirements of the ISO/IEC 23894:2023 standard, which describes the process of risks management of the artificial intelligence system.

7.2 Design Constraints

Real life constraints should be taken into account when speaking about the design of the engineering project, especially ethical, environmental, and safety concerns.

7.2.1 Ethical and Professional Responsibilities

There are certain ethical issues considered when designing the suggested framework. The algorithmic bias is minimized owing to the use of OpenDeepfake Preview dataset that includes real and fake images in equal numbers, thereby reducing the bias during binary classification. Besides, the project respects intellectual property laws, as it uses open source code and programs under licenses like MIT and Apache 2.0 and cites all the external sources used in the thesis. However, despite the fact that the suggested framework serves as a defensive deepfake detector, there are certain elements that could be misused to enhance the deepfake generation algorithms. That is why the weights of the algorithm should be distributed exclusively for the purposes of research.

7.2.2 Environmental and Sustainability Considerations

Training of large scale vision transformers (ViTs) or deep models consumes huge amounts of energy, resulting in the emission of greenhouse gases in the environment. In order to be environmentally friendly, we have developed a custom light weight encoder like the ResNet, combined with the 8-slot slot attention layer. The entire model consists of only around 2.26 million parameters, which is much lower compared to ViT models, thus reducing the number of GPU hours and energy consumed. We use memory efficient data pipeline with tf.data, which helps to process the images while loading, and thus reduces the energy used by our training servers.

7.2.3 Health and Safety Considerations

Many different dangers can be associated with deepfake technology in its non consensual application, in media, fraudulent schemes, and misrepresentation. The proposed detector plays a vital role in guaranteeing the safety of media regarding social security and privacy of individuals. It is well known that deep learning algorithms work as black box models that can be very hazardous for any legal and forensic practice. The use of the Slot Attention technique in our model provides us with the maps of attention that focus on facial regions important for classification.

7.3 Complex Engineering Problem (CEP) Coverage

In this part, it will be shown how the CEP is tackled within the context of the project, as stated by the attributes of BAETE/ABET.

Table 7.1: Mapping of Knowledge Profiles (Ks) to Project Implementation

Ks	**Knowledge Profile Attribute**	**How Ks are Addressed Through the Project**	**Evidence (Section Reference)**
K1	Natural Sciences	This is done by applying the topic within the study of facial biometrics, the principle of image formation, lighting, skin texture, and visual distortion through deepfake techniques.	Section 2.1 (Deepfake Generation and Facial Biometrics)
K2	Mathematics & Computer Science	It can be applied in the contrastive learning equation, Cosine similarity computation, NT-Xent loss reduction, statistics and machine learning algorithms.	Section 3.3 (Mathematical Foundations and Loss Functions)

Ks	Knowledge Profile Attribute	How Ks are Addressed Through the Project	Evidence (Section Reference)
K3	Engineering Fundamentals	The principles of deep learning, convolutional neural networks, principle of residual networks, activation functions (ReLu, sigmoid) and feature extraction and optimization techniques were applied to design the model.	Section 3.1 (Proposed Architecture)
K4	Specialist Knowledge	Showcased using applications of self-supervised contrastive learning, representation learning, feature modeling of objects, and Slot Attention techniques to detect deepfakes.	Section 3.4 (Contrastive Learning and Slot Attention Module)
K5	Engineering Design	Included in the formulation of the entire detection model, including encoder structure, Slot Attention mechanism, classification head, hyperparameters, and training process.	Section 3.1 (System Design), Section 3.2 (Design Methodology)
K6	Engineering Practice & Technology	Implemented with the use of implementation of programming languages such as Python and TensorFlow, TensorFlow, NumPy, PIL, Hugging Face Datasets, and many other sources like GPU accelerated training sources like NVIDIA Tesla T4s.	Section 4.2 (Development Tools and Technologies), Section 4.7 (Hardware and Software Requirements)

Ks	Knowledge Profile Attribute	How Ks are Addressed Through the Project	Evidence (Section Reference)
K7	Society, Ethics and Sustainability	Solved with the help of tackling the issue of fake news, validation of the media online, verifying the authenticity of internet media, and achieving computational efficiency.	Section 1.6 (Motivation and Social Impact), Section 7.2 (Ethical and Sustainability Considerations)
K8	Research Literature	Obtained from the analysis of the recent scientific articles on deepfake detection, contrastive learning, self-supervised learning, and object centric representation learning.	Chapter 2 (Literature Review)

Table 7.2: Mapping of Complex Engineering Problems (Ps) to Project Implementation

Ps	Complex Engineering Problem Attribute	How Ps are Addressed Through the Project	Evidence (Section Reference)
P1	Depth of Knowledge Required	Specific expertise in deep learning, self-supervised learning, representation of features, object centric learning, and computer vision is required for the project. Contrastive pre training, Slot Attention, and Deep Neural Network have been used in this project to address the issues of deepfake detection.	Section 3.1 (Proposed Architecture), Section 3.4 (Contrastive Learning and Slot Attention)

Ps	Complex Engineering Problem Attribute	How Ps are Addressed Through the Project	Evidence (Section Reference)
P2	Range of Conflicting Requirements	This suggested system comprises trade offs between various conflicting requirements like high accuracy in detection, good generalization for deepfakes that were not seen earlier, computational efficiency, and simplicity of the model.	Section 1.3 (Problem Statement), Section 3.2 (Design Considerations), Section 5.4 (Performance Analysis)
P3	Depth of Analysis Required	A thorough analysis was needed to determine the limitations of classical supervised approaches and create an architecture which can learn generalized face representations and extract manipulated areas.	Section 2.5 (Research Gap Analysis), Section 3.4 (Feature Representation Strategy)
P4	Familiarity of Issues	The technology used to generate Deepfakes is constantly improving because of GANs and diffusion models. They create new and ever changing patterns of manipulations which have to be detected using innovative methods.	Section 1.1 (Background), Section 2.2 (Deepfake Technologies), Section 5.3 (Generalization Performance)
P5	Extent of Applicable Codes	Available software libraries offer pre-existing parts of a neural network but not the architecture combining contrastive learning and Slot Attention for Deepfake recognition.	Section 4.4 (Model Implementation), Section 4.5 (Training Pipeline)

Ps	Complex Engineering Problem Attribute	How Ps are Addressed Through the Project	Evidence (Section Reference)
P6	Extent of Stakeholder Involvement	This study aims to cater to various requirements of the stakeholders such as social networking sites, digital forensics experts, researchers, policymakers, and end users having an interest in misinformation and media verification.	Section 1.6 (Social Impact), Section 7.2 (Ethical Considerations)
P7	Interdependence	The proposed framework consists of different modules which depend upon each other and consist of pre processing, contrastive pre training, feature extraction, Slot attention, classification, and evaluation. Any modification in any module will have its repercussions on the whole framework.	Section 3.1 (System Overview), Section 4.4 (Model Workflow)

7.4 Complex Engineering Activities (CEA) Coverage

This section explains how the projects involve Complex Engineering Activities (CEA).

Table 7.3: Mapping of Complex Engineering Activities (As) to Project Implementation

As	**Complex Engineering Activity Attribute**	**How As are Addressed Through the Project**	**Evidence (Section Reference)**
A1	Range of Resources	The resources employed in the development of the project include machine learning software such as TensorFlow and Keras, image processing software such as NumPy and PIL, GPU computing capability provided by NVIDIA Tesla T4 and other CUDA GPUs, and Open-Deepfake Preview Dataset.	Section 4.2 (Development Tools and Libraries), Section 4.7 (Hardware and Software Requirements)
A2	Level of Interaction	The system entails interactions between multiple system components including data preprocessing, contrastive pretraining, features extraction, Slot Attention, classification, and performance assessment. The successful integration of these modules is important for the system's overall performance.	Section 3.1 (System Overview), Section 4.4 (Model Workflow)

As	Complex Engineering Activity Attribute	How As are Addressed Through the Project	Evidence (Section Reference)
A4	Consequences for Society and Environment	The architecture described above is important when it comes to combating digital disinformation, authentication validation of media content, and the ethical use of artificial intelligence. Also, the lighter approach ensures that there are reduced computational costs when compared to heavy deep learning models.	Section 1.6 (Social Impact), Section 7.2 (Ethical and Environmental Considerations)
A5	Familiarity	The work stands out from ordinary undergraduate software engineering projects because of the use of advanced self-supervised, contrastive representation learning, and object centric attention in addressing the research challenge.	Section 2.5 (Research Gap), Section 5.3 (Experimental Analysis)

Table 7.4: Course Outcomes Mapping to this Thesis

CO	**Description**	**How Addressed in the Thesis**	**Mapped PO**	**Evidence (Report Reference)**
CO1	Real life problem identification and translation to engineering solution	This project addresses the real world issue of deepfake images and proposes an image manipulation detection mechanism using deep learning techniques based on contrastive learning and Slot Attention.	PO2, PO4	Section 1.1 (Background and Motivation), Section 1.2 (Problem Statement), Chapter 2 (Literature Review)
CO2	Identify outcomes and functional requirements with specifications	Functional requirements like image pre processing, feature extraction, manipulation detection, binary classification, and performance evaluation have been recognized and implemented.	PO3, PO2	Section 3.1 (Requirement Analysis), Section 3.2 (High-Level Architecture), Section 4.3 (Preprocessing Steps)

CO	Description	How Addressed in the Thesis	Mapped PO	Evidence (Report Reference)
CO3	Project milestones, timeline and budget planning	The project was conducted using a very well defined timetable that involved literature review, data set creation, model formulation, training, testing, evaluation and thesis writing. The cost of the project was also determined.	PO11	Chapter 6 (Project Planning and Budget), Section 6.1 (Project Timeline), Section 6.2 (Budget Estimation)
CO4	Environmental considerations and sustainability	The suggested framework adopts a light weight architecture and open source components, thus minimizing the consumption of computational resources.	PO7	Section 1.7.3 (Environmental and Sustainability Issues), Section 7.2.2 (Environmental and Sustainability Considerations)

CO	Description	How Addressed in the Thesis	Mapped PO	Evidence (Report Reference)
CO5	Professional, ethical, and social impact assessment	Such issues of ethical implications arising from the use of deepfake technology, misinformation, digital trust and media integrity are covered by this research.	PO6, PO8	Section 1.7.1 (Societal and Cultural Issues), Section 1.7.2 (Health, Safety, and Legal Issues), Section 7.2.1 (Ethical and Professional Responsibilities)
CO6	Multi-disciplinary team function	The research draws from ideas that have been developed in different fields such as computer vision, machine learning, artificial intelligence, image processing and software engineering. The collaboration with my adviser was very essential in this project.	PO9	Title Page (Authors), Acknowledgements Section

CO	Description	How Addressed in the Thesis	Mapped PO	Evidence (Report Reference)
CO7	Analyze, design, build, and evaluate engineering system	Complete deepfake detection system analysis, design, development, training, testing, and evaluation was done using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.	PO3, PO5, PO4	Chapter 3 (System Analysis and Design), Chapter 4 (Methodology and Implementation), Chapter 5 (Results and Evaluation)
CO8	Technical documentation and oral presentation	The results of the project along with methodologies followed and experiments performed have been reported in thesis report and through presentations and viva sessions.	PO10, PO12	Complete Thesis Report (Chapters 1–8), Final Defense Slides & Oral Viva

7.5 Chapter Summary

In this chapter, the standards, design constraints and engineering aspects connected with the proposed framework for deepfake image detection have been discussed. The necessary software and hardware standards have been chosen to ensure the successful creation, realization and assessment of the system. The software technologies used in the project are popular, machine learning libraries, and GPU capabilities which satisfy the modern

standards of artificial intelligence and deep learning. Design constraints in terms of ethics, environment and health and safety have been considered. The design constraints in the field of ethics were directed at the problem of struggle against disinformation and maintenance of authenticity of information, as well as appropriate application of artificial intelligence technologies. The environmental design constraint was satisfied by the development of computationally effective architecture and use of open-source materials. Health and safety design constraint was ensured by minimization of possible threats connected with the inappropriate use of deepfake images. In addition, the current chapter revealed how the proposed system satisfies Complex Engineering Problem (CEP) characteristics due to high level of knowledge application, problem analysis, system design, and research based investigation. The inclusion of Complex Engineering Activities (CEA) has also been ensured by means of the employment of advanced engineering instruments, problem solving, resource use, and social factors.

Chapter 8

Conclusion and Future Work

This chapter is intended to highlight the whole thesis project and the general outcome of the research, to mention the main results, to outline the contribution of the research and the current limitations of this research and to give suggestions for further research on digital media authentication.

8.1 Summary of the Work

The main goal of this thesis was to develop a deep learning framework for deepfake image detection which was designed, implemented and generalized. The traditional supervised CNN tends to learn shortcuts memorizing the texture of the background features found in the training data, camera noise, compression artifacts and so on and fails to generalise to novel manipulation techniques.

To overcome these shortcomings, we introduced a two stage deep learning architecture of self supervised representation learning and an object centric Slot Attention mechanism. During this first step, the custom model of ResNet inspired convolutional encoder was pretrained using a self supervised contrastive learning method and the Normalized Temperature scaled Cross Entropy (NT-Xent) loss on the OpenDeepfake Preview dataset (19,999 images) that was balanced. This pre-training helped the encoder to acquire robust and generalizable representation features for facial boundaries and contours without manual annotation.

During the second stage, an attention classification head was added to the pre-trained slot

attention representations, and optimized all over. The model placed feature maps in a fixed space, so that it could pay more attention to the local components of the face in the image (such as eyes, nose, mouth, and blending artifacts at the boundaries with other patterns), which improved the face classification accuracy and transparency.

8.2 Key Findings and Contributions

The experimentally obtained results have proven a number of important conclusions about the introduced framework. First, the developed model showed the test accuracy of 90.73% on the OpenDeepfake Preview test dataset while the baseline supervised CNN model reached only 88.00% accuracy. Additionally, the self-supervised contrastive pre-training process allowed the encoder to capture generalized representations of facial features instead of learning dataset specific characteristics. Thus, the hypothesis (H1) regarding the efficiency of contrastive representation learning and its ability to ensure better generalization was confirmed. Moreover, the introduced Slot Attention successfully divided the global convolutions of the feature maps into eight slots enabling the model to pay attention to the manipulations on faces (such as eyes blending or lips boundaries) instead of the background. Finally, the proposed framework proved to be computationally efficient since it has around 2.26 million.trainable parameters while the ViT-based models have up to 86 million parameters.

8.3 Limitations of the Study

Despite the promising performance of the proposed framework, there are several limitations that have been found during this research. First, this research considers deepfake image forgery detection and ignores the temporal inconsistency and video artifacts in the deepfake video forgery. Second, due to computational limits of the local NVIDIA MX350 GPU and the cloud-based NVIDIA Tesla T4 GPU, the model was trained for (128×128) and (224×224) pixel resolutions, which can hinder the detection of fine resolution forgery artifacts. In addition, the training and fine tuning of the model have been done only using the OpenDeepfake Preview dataset. While it is diversified, it still lacks cross dataset evaluation with benchmarks like FaceForensics++ and Celeb-DF datasets. Finally, while the

second stage of the model is computationally cheap, the self-supervised pretraining stage is computationally expensive since it requires big batch size and many training epochs to learn robust features.

8.4 Recommendations and Future Work

There are some possible directions for future studies derived from the current findings, limitations, and gained insights. Firstly, the proposed ResNet Slot Attention approach could be developed to identify deepfake videos using different techniques that allow modeling temporal inconsistencies, such as artificial facial motions and blinking, like LSTM, GRU, or Temporal Transformers. Secondly, the model's generalization potential could be studied through cross benchmark analysis by training the network using one dataset (Celeb-DF) and applying it to other popular benchmarks, such as DFDC. Moreover, there are several potential research ideas related to multi-scale Slot Attention capable of handling high resolution images ((512×512) and (1024×1024) pixels), which could be used to recognize small forgery artifacts, for example, color boundary inconsistencies or camera sensor noise patterns. Finally, the robustness of the proposed approach could be increased by analyzing the model's performance under various adversarial attacks on Slot Attention maps and developing adversarial training approaches.

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Appendices

Appendix A: Source Code / GitHub Link

The codes, scripts, pre processing modules, and evaluation files used in this study have all been released to the public domain on the following GitHub link.

https://github.com/NafisaAli-UX/Deepfake-Image-Detection-Using-Self-Supervised-Contrastive-Learning-and-Slot-Attention-

Appendix B: Team Responsibilities

This project was carried out by the collaboration of all the team members whose roles are as described below:

Sumaya Jahan Othye

- **Area of Focus:** Model Architecture Design, Self-Supervised Contrastive Pretraining, Slot Attention Integration, and Code Implementation.
- **Contributions:** Conceived the primary design of combining self-supervised contrastive pre-training with Slot Attention. Developed and implemented the custom ResNet encoder, projection head, and the 8-slot Slot Attention module in TensorFlow. Handled dimensional consistency of feature maps and slot matrices. Primary author of Chapter 3 (System Analysis & Design) and Chapter 4 (Methodology and Implementation).

Nafisa Ali

- **Area of Focus:** Dataset Management, Preprocessing Pipeline, Baseline CNN Implementation, and Training Optimization.
- **Contributions:** Managed the acquisition, cleaning, and preprocessing of the OpenDeepfake Preview dataset consisting of 19,999 images. Designed the data augmentation pipeline (rotation, flipping, contrast adjustments) and supervised the training execution using GPU instances. Implemented and evaluated the baseline CNN model for comparative performance analysis. Handled hyperparameter tuning and model optimization logs. Author of Chapter 1 (Introduction), Chapter 6 (Project Planning & Budget), and Chapter 7 (Standards & Constraints).

Mahia Akter Momo

- **Area of Focus:** Model Evaluation Recording and Report Formatting.
- **Contributions:** Recorded the evaluation metrics (accuracy, precision, recall, F1, and ROC-AUC) during the test phase of the model. Generated the performance tables and comparison graphs. Formatted and edited Chapter 5 (Results & Evaluation) and Chapter 8 (Conclusion).

Appendix C: Ethics and Compliance Statements

The following is the code of conduct and ethics principles for the current scientific investigation and software development project:

- **Data Ethics of Biometrics:** The dataset being used in this project OpenDeepfake Preview consists of facial images that were extracted from research data sources available publicly. None of the biometric data was collected, stored, or distributed to the third parties. The image frames which are used for training of the models are stored in memory temporarily only.

- **Responsible Use of Artificial Intelligence:** The deepfake detector suggested by the team can be used solely to authenticate the media files, protect from cyber attacks, and do digital forensics work. Neither the models nor the weights or algorithms will be used to generate any kind of fake media content.
- **Academic Honesty:** Any libraries or references have been cited properly. We confirm that all the material presented in this report is the property of the authors only and is not plagiarized.

Appendix D: Evidences of CEP

Attributes of CEP characteristics mentioned in Table 7.1 are illustrated by the following:

- **P1 (Depth of Knowledge) and P3 (Depth of Analysis):** Shown in Section 3.3 (Mathematical Formulations), where detailed calculations of NT-Xent loss, Cosine Similarity, and Slot Attention query key routing equations are shown.
- **P2 (Conflicting Requirements):** Shown in Section 3.2 (Design Alternatives), where the trade off analysis is described regarding parameters size, VRAM restrictions, and detection accuracy of heavy Vision Transformers compared to our proposed 2.26 million parameter network architecture.
- **P5 (Extent of Codes):** Illustrated in Section 4.5 (Model Development), where special code fragments are shown that were developed to integrate TensorFlow encoder output shapes into Slot Attention matrices.

Appendix E: Evidences of CEA

Validation of Complex Engineering Activities (CEA) features as listed in Table 7.2 can be done using:

- **A1 (Range of Resources):** All data related to multi stack (TensorFlow, TensorFlow, Python) and image dataset (includes 19,999 images) is available in chapter 4.2 (Tools

& Libraries), and all data related to the hardware (Tesla T4, MX350) is given in chapter 4.7 (Hardware).

- **A2 (Level of Interaction):** Demonstrated in Section 4.4 (Model Training) that includes the scheduling and balancing the learning rates of the self-supervised contrastive pretraining stage and the downstream supervised fine tuning stage.